# ARBITER: Supplementary material

## Contents

Methods appendix · Supplementary results · Prompt templates and example · Run provenance
eTable 1. AVATAR complete trace
eTable 2. Benchmark trials span sixteen meta-analyses
eTable 3. Published label distributions by domain
eTable 4. Confusion matrices against Published
eTable 5. Per-meta-analysis weighted κ
eTable 6. Agreement against Published, all sources and domains
eTable 7. Paired differences against Published
eTable 8. Post hoc: unweighted agreement with published labels is modest
eTable 9. Adjudicators usually agreed with models when they disagreed with Published
eTable 10. Published labels were usually less severe than the agreed choice
eTable 11. Preferred shares among adjudicated arm disagreements: sensitivities
eTable 12. Post hoc: preferred shares among adjudicated arm disagreements by strictness
eTable 13. Independent and selected ratings: full agreement including AC2
eTable 14. Localisation population eligibility
eTable 15. Error-localisation classes, both populations
eTable 16. Questions that alone changed the rating: frequencies, both populations
eTable 17. Unresolved independent answer
eTable 18. Exploratory signalling-answer agreement
eTable 19. Post hoc: supplied documents and answers in question 5.1 losses
eTable 20. Post hoc: rules-only changes make three arm disagreements ties
eTable 21. RoB 2 consistency and structural audit, all sources and domains
eTable 22. Required question skipped sensitivity
eTable 23. Quotation syntax in active justifications
eTable 24. Unassisted justification content
eTable 25. Time, cost and token use per trial by model
eTable 26. API cost reconstruction: estimates and bounds
eTable 27. Token reconciliation and compaction diagnostics
eTable 28. Evidence and quotation locations for the prior-work comparison
eTable 29. Agreement against the composite reference, Sonnet only
eTable 30. Paired differences against the composite reference
eTable 31. Composite-reference sensitivity estimates by adjudicator
eTable 32. Composite-reference paired differences by adjudicator
eFigure 1. Agreement with Published varied across meta-analyses

## Methods appendix

### Algorithms and validation

The official 22 August 2019 RoB 2 algorithms were implemented for D1 randomisation, D2 deviations (effect of assignment), D3 missing outcome data, D4 outcome measurement and D5 selection of the reported result. Numbered questions map to the 22 exported signalling questions. Required NA or missing answers make a nonempty path invalid; absent labels or wholly absent answer sets are not evaluable. Overall is the worst domain judgement, with High permitted when at least two domains have Some concerns. The guidance was cross-checked with ARBITER 1.0.0 source (commit 2f549ae). Unit tests cover the decision paths and required-answer checks. Bundle proposals reproduce 510/510 domain outputs; no overrides were recorded. All required active answers in the ARBITER domain histories are present. Unassisted invalid-path sensitivity recursively resolves required invalid answers as N/PN, then NI; off-path answers are not imputed.

### Input parity

Both arms received each trial folder as delivered, including the main article, supplements and sources list. For the original 105-trial runs, the Unassisted inputs also included the exact Markdown projection of the registry records shown in the ARBITER run (74 trials), and the 60 protocol and statistical analysis plan PDFs acquired from ClinicalTrials.gov. These inputs were copied from the ARBITER run; nothing was fetched again. For De Waal 2021, the registry record supplied to both arms belongs to a different trial. This shared input error means that registry availability does not establish relevance to the assessed trial. Source: no-kit-sonnet-5.5/README.md.

**API list-price reconstruction**

Costs are frozen-rate API list-price equivalents, not expenditure. USD rates per million tokens are: Sonnet input 2.00, 1-hour cache write 4.00, 5-minute write 2.50, cache read 0.10, output 10.00; Rates were frozen from the provider pricing documentation on 18 October 2026. Main-thread assistant events are deduplicated by message identity within turn; synthetic zero-usage events are excluded. Thinking tokens are included in output. Side calls are total recorded model usage minus terminal usage. One-hour side-call cache writes are primary; five-minute writes are a sensitivity. Reconstruction bounds coincide. Routing was global, with no geography multiplier. For each trial, equivalent US$ = (2 × input + 4 × 1-hour cache writes + 2.5 × 5-minute cache writes + 0.1 × cache reads + 10 × output tokens) / 1,000,000, summed over recorded model calls. Runs used a subscription; these equivalents are not subscription expenditure. Harness costs are unusable: unrecognised model and unknown cost basis; retained only as supplementary diagnostics.

**Error localisation**

Losses are cells where both adjudicators agreed where a source label differs from preferred. A reference adjudicator is usable only when their independent judgement equals the adjudicators' agreed choice. Primary requires both; sensitivity at least one. Y/PY collapse to Y and N/PN to N; NI and NA remain separate. Discordance requires all usable adjudicators to agree on a parsed answer different from the losing answer. Adjudicator disagreements are separately flagged, never errors. A single replacement is decisive if it changes a valid algorithm output, even if the new output is not preferred. Other answers, including inactive NA, stay fixed; invalid substituted paths are not decisive. First applicable cell class: judgement departs from own answers; one question alone changed the rating; several questions each changed the rating alone; jointly decisive (all differing replacements reach preferred); no differing questions with complete comparisons; not attributable. The no-discordance class can contain adjudicator-disagreement questions. Reference-judgement departures from their own answers are flagged. Frequencies count all questions that alone changed the rating regardless of cell class; joint sets contribute none. Ambiguous or qualified answers are not imputed. Published losses cannot be localised without signalling answers.

**Evidence binding and verification**

Each distinct evidence identity cited in bases or counterevidence by current active answers is counted once per bundle; superseded histories are excluded. Visual records (figure kind or render field) take precedence over text spans. Text spans (narrative kind or quote/start/end fields) require a nonblank quote and source version, source identity matching a captured bundle source, finite page/start/end, and start less than end. Quotes are server-extracted spans cited by handle. Visual binding requires render source matching the record and a captured source, positive integer page, nonblank PNG hash, four finite region coordinates in [0,1] with ordered corners, and nonblank transcription. Images are server-rendered and hash-bound; transcriptions are model-written and are not machine-checked against the image. Unknown or missing records are other and unbound in the overall denominator. Empty bundles contribute no cited records. Trials using visual evidence use all 102 trials as denominator. These are structural checks, without PDF re-extraction or a judgement of whether a passage supports an answer. Bundle verification does not establish clinical correctness or completion. Unassisted evidence traces are not available.

**Inference and references**

All estimates use the fixed 102-trial set. CIs use 10,000 trial-cluster bootstrap replicates stratified by meta-analysis, percentile intervals (quantile type 7), seed 20261018; the same weights support paired differences. Missing predictions count as wrong. For ordinal coefficients a missing prediction receives maximal disagreement; the same rule applies to all estimates. Independent signalling-answer agreement uses the earlier exploratory Y/PY, N/PN and NI/NA collapse. Higgins JPT, Savović J, Page MJ, Sterne JAC, editors. Revised Cochrane risk-of-bias tool for randomized trials (RoB 2), 22 August 2019.
https://www.riskofbias.info/welcome/rob-2-0-tool/current-version-of-rob-2. Salman A et al. ARBITER 1.0.0, public source release, commit 2f549ae.
https://github.com/AliSalman-et-al/ARBITER/tree/2f549ae30f056ff441ea61a04b719968716f2219. Anthropic. Claude API pricing. Accessed 18 October 2026. https://platform.claude.com/docs/en/about-claude/pricing.

**Published-label target matching**

LABEL-TARGETS.csv links each published label to its review, source table, outcome, time point and contrast, with quoted binding evidence where stated and an explicit inference otherwise. The assessment prompt targets the labelled result rather than automatically the trial primary outcome, with the definition, printed result and locator from the trial report. Among the 102 included trials, 33 targets match an explicitly stated result, 11 match one of several covered outcomes, and 32 match an inferred result; 7 partly match, 9 differ and 10 have unknown binding. These categories are retained rather than treating all targets as exact matches. Twelve published label sets (60 domain labels) cover every outcome taken from a trial; another 10 label sets (50 domain labels) have no identified result, and seven more label sets (35 domain labels) jointly cover pain and disability; thus 29 sets (145 domain labels) do not isolate a single result (19 broad and 10 unknown). Binding was inferred rather than

explicitly stated for 62 label sets, overlapping these categories. Source: benchmark README-FOR-ALI.md, LABEL-TARGETS.csv and per-review BINDING.md; label-target-summary.csv recomputes matching categories.

**Dates and training exposure**

Trial reports span publication years 1996–2025; the 16 source reviews were published in 2025–2026. Years come from TRIAL-SUMMARY.csv and trial-label years where that field is blank; the benchmark README identifies the SANCTUARY report as 2025. Anthropic documents June 2026 as the training-data cut-off for claude-sonnet-5-5 (model documentation, https://platform.claude.com/docs/en/models/sonnet-5-5/overview, and provider help, https://support.claude.com/en/articles/8114494-how-up-to-date-is-claude-s-training-data; checked 19 October 2026). Dates cannot establish whether a report was used in training.

**Adjudication process and primary endpoint**

Two blinded adjudicators used no AI. Each rated the domain from the supplied documents and froze the independent rating before seeing any label. They then saw only the three labels, randomised in order for each cell, and chose the preferred label with a reason. They did not see model rationales or quotations. The primary endpoint is the share of adjudicated arm-disagreement cells where the adjudicators preferred ARBITER's rating. Independent-rating paired differences added post hoc compare ARBITER and Unassisted agreement with the same frozen independent rating on all 154 disputed cells, using the shared stratified trial-cluster bootstrap.

**Limitations**

Adjudication selected cells on disagreement, so preferred shares are subject to discrepant-analysis selection and do not estimate full-benchmark accuracy. Adjudicators chose only among the offered labels; cells where both arms matched Published were not audited. Labels alone may have revealed an arm, although this risk was low because rationales and quotations were hidden. Each arm ran once; workflow, prompt and runtime contributions cannot be separated. Reports or reviews may have appeared in training before the documented June 2026 cut-off. Whether quotations support the answers was not assessed. Overall was not adjudicated. Some label-target bindings were inferred, broad or mismatched; disagreement with a published label may partly reflect outcome mismatch rather than a different judgement.

## eTable 1. AVATAR complete trace

### Part A. Answers and judgements

| Question | ARBITER | Evidence handles (type) | Justification | Unknowns | Counterevidence | Unassisted |
|---|---|---|---|---|---|---|
| D1 judgement / rationale | Proposed: Low; adopted: Low | | | | | Low<br>Allocation used a computer-generated, centre-stratified permuted-block sequence delivered through a web-based interactive response system, so concealment was adequate. The baseline differences in age (P = .02) and haemoglobin (P = .01) are consistent with chance across about 25 variables. |
| 1.1 Random allocation sequence | Yes | eh_d6c269341c7d8a5d (direct support) | The original AVATAR paper states the assignment was computer-generated and stratified by centre using a permuted-block sequence with variable block size, which is a random component. | — | — | Yes |
| 1.2 Concealment until enrolment and assignment | Yes | eh_d6c269341c7d8a5d (direct support) | Patients were randomized using a web-based interactive response system, a central remote allocation method independent of enrolment staff. Variable block size limits predictability of the last allocations in a block. | Whether the central system was run by a unit independent of enrolling staff is not stated explicitly. | — | Yes |
| 1.3 Preintervention imbalance suggesting a randomisation problem | No | eh_92eed84c7b98e37c (direct support); eh_58abdac52262aa33 (context) | Group sizes were 78 vs 79. Of about 25 baseline comparisons in Table 2, only age (P=.02) and haemoglobin (P=.01) were nominally significant, and diabetes was P=.07. The age difference is 1 year in medians. This is compatible with chance and none approaches P<0.001. | — | Basis 1: Diabetes was less common in early surgery (17.9% vs 29.1%), which could favour surgery, but P=.07 is compatible with chance and the 2022 paper attributes it to chance. | Probably no |
| D2 judgement / rationale | Proposed: Low; adopted: Low | | | | | Low<br>The trial was open label, so participants and clinicians knew the assignments. Surgery in the conservative arm followed |

| Question | ARBITER | Evidence handles (type) | Justification | Unknowns | Counterevidence | Unassisted |
|---|---|---|---|---|---|---|
| | | | | | | protocol-defined triggers such as symptom onset. Six of 78 early-surgery patients did not receive surgery, and some surgeries were delayed by COVID-19. These are not deviations likely to affect the outcome. The analysis was intention-to-treat. |
| 2.1 Participants aware of assignment | Yes | eh_975aac3202107531 (direct support); eh_3ce4199af9d487f3 (context) | The registry lists masking as NONE, and the comparison is surgical valve replacement versus watchful waiting, which cannot be hidden from participants. | — | — | Yes |
| 2.2 Carers or intervention providers aware of assignment | Yes | eh_975aac3202107531 (direct support) | Open-label surgery versus conservative care means clinicians and surgeons delivering care knew the assignment. Allocation was concealed until assignment, but blinding afterwards was impossible. | — | — | Yes |
| 2.3 Deviations arising from trial context | Probably no | eh_c8e0958840636206 (direct support); eh_22d01d9384e1a677 (context); eh_4b5761bf573cc287 (context); eh_05e667d2d41d3858 (context) | Crossover from conservative care to surgery (35/79) was driven by protocol-defined indications (symptom onset, AS progression, LVEF decline), as Table 3 and the protocol show, so it is consistent with the protocol rather than trial-context driven. Surgical delays beyond 8 weeks arose from the COVID-19 public-health emergency and were documented as an anticipated minor deviation in the SAP amendment. Four early-surgery patients did not receive surgery as assigned, for reasons like infection or declining, which could occur outside a trial. | Reasons for declining surgery are not described in detail. | Basis 3: The authors note pandemic-related follow-up substitution by phone and lengthened AVR delays could have increased clinical risk. These arose from the pandemic, not the trial context, and no deaths occurred while awaiting scheduled surgery in the conservative group. | Probably no |
| 2.4 Deviations affecting the outcome | Not asked (inactive; branching) | | | | | Not applicable |
| 2.5 Deviations balanced across groups | Not asked (inactive; branching) | | | | | Not applicable |
| 2.6 Appropriate analysis of | Yes | eh_30fa222ec5523dcc (direct support); | All analyses were intention-to-treat as randomized, and all 157 patients were included. One early-surgery | — | — | Yes |

| Question | ARBITER | Evidence handles (type) | Justification | Unknowns | Counterevidence | Unassisted |
|---|---|---|---|---|---|---|
| assignment effect | | eh_c8e0958840636206 (direct support) | patient lost to follow-up was censored and still analysed. | | | |
| 2.7 Substantial impact of analysing outside assigned groups | Not asked (inactive; branching) | | | | | Not applicable |
| D3 judgement / rationale | Proposed: Low; adopted: Low | | | | | Low<br>All 157 patients were analysed. One patient lost to follow-up was censored. |
| 3.1 Outcome data available for nearly everyone | Yes | eh_e66d374e838e8c6d (direct support) | All 157 randomized patients were analysed and median follow-up was 63 months. Only one patient (early-surgery arm) was lost to follow-up and censored at last contact, so 1/157 (0.6%) has incomplete outcome data. This is far fewer than the observed deaths, so any outcome for that patient could not materially change the HR of 0.44. | — | — | Yes |
| 3.2 Evidence against bias from missing data | Not asked (inactive; branching) | | | | | Not applicable |
| 3.3 Missingness potentially depends on true outcome | Not asked (inactive; branching) | | | | | Not applicable |
| 3.4 Missingness likely depends on true outcome | Not asked (inactive; branching) | | | | | Not applicable |
| D4 judgement / rationale | Proposed: Low; adopted: Low | | | | | Low<br>Death is an objective outcome. Assessors were not blinded to allocation, but this cannot plausibly bias an all-cause death count. |
| 4.1 Inappropriate measurement method | Probably no | eh_3ce4199af9d487f3 (direct support) | All-cause death is a pre-specified, objective event ascertained through scheduled follow-up (6-monthly in year 1, then yearly), phone contact when in-person visits were not possible, and review of medical records for any registered event. Nothing suggests an insensitive method or an instrument with poor validity. | — | — | No |
| 4.2 Measurement | Probably no | eh_3ce4199af9d487f3 | Both arms followed the same protocol schedule, and | Whether vital | — | No |

| Question | ARBITER | Evidence handles (type) | Justification | Unknowns | Counterevidence | Unassisted |
|---|---|---|---|---|---|---|
| differs across groups | | (direct support) | COVID-related phone follow-up applied to patients regardless of arm. Death is hard to miss, so any difference in contact opportunities (e.g. surgery visits) is unlikely to change ascertainment of death. | status was verified from registries in both arms is not stated. | | |
| 4.3 Assessors aware of received intervention | Yes | eh_3ce4199af9d487f3 (direct support) | The trial was open-label (registry masking: none), and the report states that DSMB members were not blinded to allocation when reviewing events. Site investigators recording deaths were also aware of the allocation. | — | — | Not applicable |
| 4.4 Assessment potentially influenced by awareness | Probably no | eh_3ce4199af9d487f3 (inference) | All-cause death is an observer-reported outcome that involves no judgement, so knowledge of the intervention is unlikely to influence its assessment, even though the assessors were aware of the allocation. | — | — | Not applicable |
| 4.5 Assessment likely influenced by awareness | Not asked (inactive; branching) | | | | | Not applicable |
| D5 judgement / rationale | Proposed: Some concerns; adopted: Some concerns | | | | | Low<br>Time to death was a prespecified secondary endpoint with a registered 5-year time frame. It was analysed with a single Cox model, and the SAP amendment only addresses COVID-19 delays. I found no sign of selective reporting. A small concern is that the extended follow-up was a later analysis, but it matches the registered time frame. |
| 5.1 Analysis plan finalised before unblinding | No information | eh_d54a2600ddc753b3 (context); eh_7ad155913c1c5826 (context); eh_30fa222ec5523dcc (context); eh_7961ea6680167af2 (context) | The registry entry (first submitted 30 Apr 2015, before enrolment began in June 2015) names all-cause death only as a component of composite outcomes. It does not list time to all-cause death as a separate outcome. The Circulation 2022 and EHJ 2024 reports describe 'time to death' as a prespecified secondary end point analysed with Kaplan-Meier and Cox models, citing a published protocol. That protocol is not among the captured Sources, and the report statements are not date-stamped before unblinded | Whether the published protocol listed time to all-cause death and its Cox analysis, and its date relative to unblinded data. | — | Probably yes |

| Question | ARBITER | Evidence handles (type) | Justification | Unknowns | Counterevidence | Unassisted |
|---|---|---|---|---|---|---|
| | | | data. The only SAP-type document is an amendment dated 4 Jan 2021 about COVID surgical delays. No dated source therefore specifies this measurement and analysis. | | | |
| 5.2 Result selected from eligible outcome measurements | Probably no | eh_30fa222ec5523dcc (direct support); eh_d54a2600ddc753b3 (context) | All-cause death is a single, unambiguous event, so there is essentially one way to measure the outcome domain. The registered time frames were 36 months and 5 years. The 2024 report states that all-cause death was analysed over the full follow-up and also restricted to 3 years, so no eligible time point appears to have been omitted in favour of a more favourable one. | — | — | Probably no |
| 5.3 Result selected from eligible analyses | Probably no | eh_30fa222ec5523dcc (direct support); eh_f625637d366df2a0 (context) | Time-to-event secondary outcomes were analysed with the same Kaplan-Meier, log-rank and Cox model (treatment only) used for the primary outcome, and the same method was used in the 2022 paper. No covariate-adjusted or alternative models are described, and the long-term and 3-year analyses are both reported. | — | — | Probably no |
| Overall | Proposed: Some concerns; adopted: Some concerns | | Any domain Some concerns, with no High domain: Overall Some concerns. | | | Low |

## Part B. Evidence records

| Handle | Source | Page | Lines | Source-version hash | Full verbatim quote |
|---|---|---|---|---|---|
| eh_05e667d2d41d3858 | ehae585.pdf | 9 | 76–83 | a890f771fd05 | follow-up and lengthening AVR delays. Pre-defined physical visits were substituted by phone follow-up. Such surveillance within the watchful waiting may hypothetically impact timely detection of symptoms conse quently leading to increased clinical risk, including sudden death. Nevertheless, in the conservative group, none of the patients experi enced MACE or died while awaiting the scheduled surgery. It is of note that in the small number of patients undergoing TAVR, delays ap peared shorter as compared with the surgical intervention. |
| eh_22d01d9384e1a677 | Current registry-linked sap: SAP_000.pdf | 2 | 1–12 | ba0a93294998 | Amendmen to statystical analysis 1. Patients randomized to AVR surgery who received delayed surgery (more than 8 weeks after randomization) due to COVID -19 public health emergency will be documented and this delay will be defined as an anticipated minor deviation from the clinical protocol 2. Appropriate statistical |

| Handle | Source | Page | Lines | Source-version hash | Full verbatim quote |
|---|---|---|---|---|---|
| | | | | | procedures will be performed and results will be published according to the FDA guidance "Conduct of Clinical Trials of Medical Products During the COVID-19 Public Health Emergency Guidance for Industry, investigators, and Institutional Review Boards" |
| eh_30fa222ec5523dcc | ehae585.pdf | 5 | 13–24 | a890f771fd05 | Statistical analysis We conducted all analyses according to the intention-to-treat principle, where all patients were included in the analysis as they were randomized. For the analysis of the primary outcome, we built a Kaplan-Meier graph with a log rank P-value to test for a difference between early surgery and control intervention at a two-sided α value of 0.05 and used a Cox propor tional hazards model to derive hazard ratios (HRs) with 95% confidence in tervals (CIs) for treatment effect estimation. Time-to-event secondary outcomes were analysed using the same meth ods applied for the analysis of the primary outcome. For MACE, all-cause death, HF hospitalization, and cardiovascular death, we repeated the ana lysis as described above restricted to a 3-year follow-up, with complete |
| eh_3ce4199af9d487f3 | ehae585.pdf | 3 | 1–116 | a890f771fd05 | Introduction Aortic stenosis (AS) is the single most common valvular heart disease re quiring intervention in developed countries and is projected to double by 2050 in both the USA and Europe.1 The approach to symptomatic pa tients with severe AS is straightforward. Surgical aortic valve replacement (AVR) and, more recently, transcatheter aortic valve implantation (TAVI) procedures are Class I recommendation to relieve symptoms and im prove survival in symptomatic patients with severe AS.2,3 Conversely, the question of when and how to treat truly asymptom atic patients with severe AS and normal left ventricular (LV) systolic function is still subject to debate and ongoing research. To date, two randomized trials have demonstrated benefit of early AVR in compari son with conservative strategy in asymptomatic patients with severe and very severe AS patients.4,5 In addition, there are several ongoing randomized trials investigating the role of early TAVI and/or early sur gical AVR in asymptomatic AS. The Aortic Valve Replacement Versus Conservative Treatment in Asymptomatic Severe Aortic Stenosis (AVATAR) trial is an investigator- initiated, prospective, multinational, randomized, controlled, parallel-group, event-driven trial that compared the safety and efficacy of early surgery in the treatment of truly asymptomatic patients with severe AS and LV ejec tion fraction (LVEF) ≥ 50% vs. conservative treatment with watchful wait ing strategy and AVR only after symptom onset, or other guidelines directed Class I or II indications for AVR.3,6 In the primary analysis of the AVATAR trial, asymptomatic AS patients randomized to early surgery had a lower incidence of the composite primary outcome comprising all- cause death, acute myocardial infarction, stroke, or unplanned hospitaliza tion for heart failure (HF), compared with patients who were randomized to conservative treatment.5 We now report the results of the AVATAR trial after extended follow-up at 5 years. Methods Study design and oversight The AVATAR trial (ClinicalTrials.gov identifier NCT02436655) protocol and protocol update were designed by the first co-authors and steering committee co-chairmen and was approved by the steering committee of the AVATAR trial.7,8 The clinical and outcomes data were collected by per sonnel at the participating sites and were sent directly to the data coordinat ing centre at the University Clinical Center of Serbia. No extramural funding was used to support this work. The authors, members of the steering com mittee, and investigators are solely responsible for the design and conduct of this trial, all analyses, drafting and editing of the article, and its final contents. Independent statistician conducted the long-term data analyses. The trial was conducted in accordance with the Declaration of Helsinki. All participants provided written informed consent. An independent data and safety monitoring |

| Handle | Source | Page | Lines | Source-version hash | Full verbatim quote |
|---|---|---|---|---|---|
| | | | | | board (DSMB) adjudicated all serious adverse events and oversaw the safety of the trial. The article was prepared by the first author and steering committee co-chairmen and was reviewed and edited by mem bers of the steering committee and authors. All authors reviewed the article, approved its submission for publication, and vouched for the accuracy and completeness of the data and for the fidelity of the trial to the protocol. Study population The study inclusion and exclusion criteria are shown in Table 1; trial flowchart is shown in Figure 1. Briefly, a total of 157 patients with severe AS according to the guidelines were enrolled in the trial: 79 were randomized to conservative treatment and 78 were randomized to early surgery. To prove truly asymp tomatic status, exercise testing was performed in all candidates to evaluate symptom status according to a standardized protocol using treadmill or semi- supine ergo-bicycle. To consider exercise testing negative, which was manda tory for inclusion, all candidates needed to reach a projected submaximal heart rate. Positive exercise test included onset of AS-related symptoms, fall in systolic blood pressure (≥20 mmHg from the baseline values), or Electrocardiography (ECG) or stress echocardiography signs of myocardial ischaemia.9 Follow-up and trial endpoints Patients were followed according to the protocol every 6 months for the first year and then yearly with the in-person visits at the participating study centre. As the trial was affected by the Covid-19 pandemic, contact with patients who could not come in person was made by phone. For any event that was registered, the medical records were asked for and reviewed. Adverse clinical events were adjudicated by the DSMB per protocol defini tions.7,8 Data and safety monitoring board members were not blinded to the treatment allocation during events review. They adjudicated the events by consensus. Table 1 Eligibility inclusion/exclusion criteria Inclusion criteria • Men and women of any ethnic origin aged ≥18 years • Written informed consent • Vmax across the aortic valve > 4 m/s or P mean ≥ 40 mmHg and AVA ≤ 1 cm2 or AVAi ≤ 0.6 cm2/m2 at rest • Without reported symptoms • Society of Thoracic Surgeons score < 8% Exclusion criteria Participation in another clinical trial within 30 days prior to randomization • Pregnant or nursing women • Mental condition rendering the patient unable to understand the nature, scope, and possible consequences of the study or to follow the protocol • Positive stress-test defined as follows: (a) Anginal chest pain during testing (b) Syncope or dizziness during testing (c) Decrease in systolic blood pressure during exercise ≥ 20 mmHg (d) Malignant arrhythmia during exercise testing (VT or VF) • Left ventricular ejection fraction < 50% at rest • Very severe AS defined as Vmax > 5.5 m/s at rest • Significant disease of other valves: mitral stenosis with Pmean > 5 mg, or any significant regurgitation ≥ 3+ • Recent previous myocardial infarction (<1 year) • Need for additional aortic root replacement (i.e. Bentall) or ascending aorta surgery in asymptomatic patients undergoing AVR • Previous coronary bypass surgery • Previous any heart valve surgery • Impaired renal function, i.e. creatinine > 200 µmol/L or glomerular filtration rate < 30 mL/min/1.73 m2 • Significant pulmonary hypertension at rest (systolic pulmonary artery pressure > 50 mmHg) • Uncontrolled hypertension at rest (systolic blood pressure > 180 mmHg and diastolic blood pressure > 100 mmHg) • Significant comorbidity with reduced life expectancy (<3 years) • Uncontrolled diabetes mellitus (HbA1c > 9%) • Significant chronic obstructive pulmonary disease (FEV1 < 70% of predicted value) |
| eh_4b5761bf573cc287 | CIRCULATIONAHA.121.057639.pdf | 4 | 8–15 | 592b9a6c531c | Patients in the conservative treatment group were referred for surgery in case of onset of AS-related symptoms, if the LVEF decreased to <50%, or if the peak aortic jet velocity increased each year by >0.3 m/s on follow-up echocardiog- raphy, according to European Society of Cardiology guide- lines on valvular |

| Handle | Source | Page | Lines | Source-version hash | Full verbatim quote |
|---|---|---|---|---|---|
| | | | | | disease at the time the study was designed.11 Conservative treatment included the treatment of risk factors and comorbidities. |
| eh_58abdac52262aa33 | ehae585.pdf | 5 | 47–50 | a890f771fd05 | Diabetes mellitus 14 (17.9%) 23 (29.1%) .07 |
| eh_7961ea6680167af2 | Current registry-linked sap: SAP_000.pdf | 1 | 1–8 | ba0a93294998 | Avatar trial protocol -Amendment to surgical delays- Aortic Valve Replacement Versus Conservative Treatment in Asymptomatic Severe Aortic Stenosis (AVATAR trial) NCT number : NCT02436655. Date of amendment: 04.01.2021 |
| eh_7ad155913c1c5826 | CIRCULATIONAHA.121.057639.pdf | 4 | 16–35 | 592b9a6c531c | Trial End Points The primary end point was a composite of all-cause mortality or major adverse cardiovascular events (MACEs) composed of acute myocardial infarction, stroke, and unplanned HF hospital- ization needing intravenous treatment with diuretics or inotro- pes. Prespecified secondary end points included the following: • In-hospital and 30-day postoperative mortality in oper- ated patients in both groups • Repeat aortic valve surgery in operated patients in both groups • Repeated MACEs, including stroke, acute myocardial infarction, or unplanned HF hospitalization needing IV diuretic treatment • Major bleeding defined as types 3, 4, and 5 according to the consensus report from the Bleeding Academic Research Consortium13 • Thromboembolic complications on the basis of clinical symptoms, signs, and imaging studies • Time to death • Time to first HF hospitalization |
| eh_92eed84c7b98e37c | ehae585.pdf | 5 | 26–136 | a890f771fd05 | Table 2 Selected baseline characteristics of the study population Early surgery group (n = 78) Conservative treatment group (n = 79) P-value Demographics and comorbidities Age, years, median (IQR) 68 (63-73) 69 (64-75) .02 Sex (female) 32 (41.0%) 35 (44.3%) .67 STS score, median (IQR) 1.6 (1.1-2.2) 1.8 (1.2-2.7) .67 Days from randomization to surgery, median (IQR) 55 (36-79) 476 (226-1098) <.001 Diabetes mellitus 14 (17.9%) 23 (29.1%) .07 Hypertension 69 (88.4%) 70 (88.6%) .44 History of PCI 1 (1.3%) 2 (2.5%) .44 History of stroke 2 (2.5%) 2 (2.5%) .92 Peripheral arterial disease 0 (0%) 1 (1.36%) .80 Laboratory parameters BNP, pg/mLa, median (IQR) 83 (53-127) 89 (58-149) .61 NT-proBNP, pg/mLa, median (IQR) 381 (153-660) 347 (186-722) .45 Haemoglobin, g/L, median (IQR) 141 (131-150) 134 (128-141) .01 Creatinine, μmol/L, median (IQR) 80 (66-94) 76 (67-92) .27 Blood glucose, mmol/L, median (IQR) 5.6 (5.3-6.7) 5.6 (5.1-6.8) .70 Echocardiography LVESV, mL, median (IQR) 28 (20-41) 33 (22-42) .96 LVEDV, mL, median (IQR) 113 (89-142) 113 (96-126) .54 LV mass index, g/m2, median (IQR) 152 (133-175) 160 (139-183) .67 Left atrium, cm 4.1 (3.8-4.4) 4.2 (3.9-4.4) .68 SVi, mL/m2, median (IQR) 39 (33-48) 42 (34-51) .58 PA systolic pressure, mmHg, median (IQR) 30 (26-36) 30 (27-37) .82 Vmax, m/s, median (IQR) 4.5 (4.3-4.8) 4.5 (4.2-4.7) .13 Pmean, mmHg, median (IQR) 51 (44-58) 50 (43-59) .16 Pmax, mmHg, median (IQR) 82 (74-90) 79 (70-90) .18 AVA, cm2, median (IQR) 0.73 (0.55-0.84) 0.74 (0.59-0.89) .29 AVAi, cm2/m2, median (IQR) 0.37 (0.3-0.42) 0.37 (0.31-0.46) .08 E/e', median (IQR) 12.2 (9.6-16.3) 12.2 (8.8-18.1) .54 |
| eh_975aac3202107531 | ClinicalTrials.gov NCT02436655 | 1 | 67–68 | d32e23eb2a1e | #### Masking info - Masking: NONE |
| eh_af1340e6c28a70eb | ehae585.pdf | 7 | 1–148 | a890f771fd05 | group was 67.9 years, while the mean age of patients in the early surgery group was 65.4 years (P = .19). Four patients in the conservative treat ment group underwent TAVI procedure. Indications for surgery in the conservative treatment group are shown in Table 3. Median time to sur gery upon establishing the indication was 123 days (IQR 90-297 days). There were no significant baseline differences between the patients in the |

| Handle | Source | Page | Lines | Source-version hash | Full verbatim quote |
|---|---|---|---|---|---|
| | | | | | conservative group who remained AVR-free during follow-up and those who later underwent AVR (see Supplementary data online, Table S1). Patients in the conservative group on the watchful waiting, in cluding those undergoing valve surgery, experienced a progressive increase in the aortic valve severity associated with a significant decline in LVEF over time (see Supplementary data online, Tables S2 and S3, respectively). In total, 57/107 (53.3%) patients received a bioprosthetic valve and 50/107 received a mechanical valve (46.7%). In the early surgery group, 34/72 (47.2%) received a bioprosthetic valve, while 23/35 (65.7%) pa tients received a bioprosthetic valve in the conservative treatment group (P = .32). Three patients in early surgery group (3/78; 3.8%) and four pa tients in the conservative treatment group (4/35; 11.4%) have undergone concomitant coronary artery bypass grafting. No peri-procedural myo cardial infarction has been noticed. The intraoperative mortality was low in both groups (1.3% in the early surgery vs. 2.5% in the conservative group; P = .54). There was no difference in length of hospitalization in the early surgery group vs. operated patients in the conservative treatment group (8.8 vs. 9.1 days; P = .87). Details of post-operative aortic valve function, procedural time, and mean hospitalization length between both groups are shown in Supplementary data online, Table S4. Three pa tients (3/78, 3.8%) in the early surgery group, and 1/35 (2.86%) patient in the conservative treatment group underwent redo AVR procedure (P = .79; Supplementary data online, Table S5. Outcomes There were a total of 55 primary outcome events comprising a com posite of all-cause mortality or acute myocardial infarction, stroke, and unplanned HF hospitalization. A primary outcome event occurred in 18/78 patients (23.1%) in the early surgery group and in 37/79 pa tients (46.8%) in the conservative treatment group (HR with early sur gery vs. conservative treatment 0.42; 95% CI 0.24-0.73, P = .002; Figure 2). Forty patients died during follow-up: 13/78 (16.7%) in the early surgery group and 27/79 (34.17%) in the conservative treatment group. In the early surgery group, 11/72 patients died after valve surgery at median of 362 days (IQR 116-1211), of which 8 were cardiovascular deaths. In the conservative group, 8/35 patients died after surgery at shorter median of 131 days (IQR 47.8-603), of which 5 were cardiovas cular deaths. Mortality causes for individual subjects are detailed in Supplementary data online, Tables S6A and S6B. Kaplan-Meier estimates of the individual endpoints of all-cause death and HF hospitalization were significantly lower in the early surgery compared with the conservative group (HR 0.44; 95% CI 0.23-0.85, P = .012, for all-cause death and HR 0.21; 95% CI 0.06-0.73, P = .007, for HF hospitalizations) (Figure 3 and Table 4). The risk of repeated MACE was significantly higher in the conservative treatment group (Table 4). Sudden cardiac death occurred in 9/79 patients (11.4%) in the con servative group compared with 4/78 patients (5.1%) in the early surgery group (P = .17; Table 4). The annual rate of sudden cardiac death was twice higher in the conservatively treated group as compared with early surgery group (2.2% vs. 1%; P = .32). If not considering symptomatic pa tients in the watchful waiting group who died suddenly at home, the an nual rate of sudden cardiac death in the conservative group reached 1.48%. One patient randomized to the early surgery group died sud denly while awaiting the surgery. There was a trend for higher rates of cardiovascular deaths in the conservative treatment group over time (19.3% vs. 11.6%; P = .1; Figure 4). The incidence of serious adverse events was also significantly higher in the conservative treatment group (49.4% vs. 26.4%; P = .013) (Table 4). No difference between groups was observed in the safety outcomes including major bleeding and thromboembolic complications (5.1% vs. 3.8%; P = .69; and 5.1% vs. 3.8%; P = .69; respectively; Table 4). |

| Handle | Source | Page | Lines | Source-version hash | Full verbatim quote |
|---|---|---|---|---|---|
| | | | | | In a post hoc heterogeneity analysis in the study population, no signifi cant interaction for heterogeneity was noted for any of the analysed parameters (see Supplementary data online, Figure S1) indicating that results are consistent across different patients' subgroups. Discussion The main findings from the extended follow-up of the AVATAR trial can be summarized as follows: (i) the primary composite outcome comprising of all-cause death, acute myocardial infarction, stroke, and HF hospitalizations was significantly and consistently lower with the early surgery approach compared with the conservative treatment ap proach; (ii) patients on conservative strategy appeared at almost 50% p = 0.01 HR = 0.44 (95% CI 0.23 to 0.85) 0% 10% 20% 30% 40% 50% 60% All-cause death incidence 79 71 61 39 14 Control 78 72 71 52 15 Early surgery Participants, n 0 20 40 60 80 Time (Months) Early surgery Conservative treatment p = 0.007 HR = 0.21 (95% CI 0.06 to 0.73) 0% 10% 20% 30% 40% 50% 60% HF hospitalization incidence 79 65 55 32 11 Control 78 71 70 51 15 Early surgery Participants, n 0 20 40 60 80 Time (Months) Early surgery Conservative treatment Figure 3 Cumulative incidence of all-cause death (upper panel) and heart failure hospitalization (lower panel) (intention-to-treat population) 4532 Banovic et al. Downloaded from academic.oup.com/eurheartj/article/45/42/4526/7742145 by guest on 03 October 2026 |
| eh_c8e0958840636206 | ehae585.pdf | 6 | 1–99 | a890f771fd05 | 3-year follow-up information available for all patients. For binary outcomes, we used a generalized linear model assuming a binomial distribution with a log link function to report risk ratios and 95% CI. We conducted subgroup analyses accompanied by a P for interaction to assess the consistency of the treatment effect on the primary outcome according to the following pre- defined variables: age (<65 vs. ≥65), sex, baseline ejection fraction (<60% vs. ≥60%), and country of trial site. P-values and 95% CI for secondary out comes and secondary analyses of the primary outcome were not adjusted for multiple testing and are considered exploratory. For the post hoc ana lyses of echocardiography readouts of valve severity and LVEF in patients in the conservative group, we calculated the mean change from baseline to 12-month follow-up for all patients in the conservative group with avail able data and the mean change from baseline to final follow-up for all pa tients in the conservative group that underwent surgery and had available data. We conducted paired t-tests to derive P-values and 95% CI for the mean changes from baseline.11 This analysis was repeated for patients in the conservative group that underwent AVR. Analyses were performed using Stata, version 18.0. Results Patients and follow-up Main patient characteristics at baseline are summarized in Table 2. The average age of enrolled patients was 67 years, 57% were men, the ma jority of patients had a degenerative aetiology of AS (84.7%), and the median estimated operative mortality according to the Society of Thoracic Surgeons predicted risk of mortality score was 1.7%. No sig nificant heterogeneity between both groups was noted. All patients were included in the analysis. Median follow-up for all patients was 63 months [interquartile range (IQR) 48-75). Median follow-up in sur vivors was 68 months (IQR 58-79). One patient, who was randomized to early surgery, has been lost to follow-up, and this patient was cen sored when the last follow-up information was available and included in the intention-to-treat analysis (Figure 1). Aortic valve replacement procedures In the early surgery group, AVR was performed in 72 of the 78 patients (92.3%). Among the six patients that did not undergo AVR per random ization, one patient, as above mentioned, was lost to follow-up, and one died prior to valve surgery. Four patients were available for the extended follow-up. Among them, one patient died unoperated after 2113 days; two became symptomatic and underwent intervention after 691 and 791 days, respectively; and one remained |

| Handle | Source | Page | Lines | Source-version hash | Full verbatim quote |
|---|---|---|---|---|---|
| | | | | | asymptomatic and AVR free throughout the entire follow-up (Figure 1). The median time from ran domization to AVR in the early surgery group was 55 days (IQR 36-79). Thirty-five out of the 79 patients (44.3%) in the conservative treatment group had surgery during follow-up. Median time from randomization to surgery in the conservative treatment group was 476 days (IQR 226-1098). At the time of AVR, the mean age of patients in the control p = 0.002 HR = 0.42 (95% CI 0.24 to 0.73) 0% 10% 20% 30% 40% 50% 60% Primary composite endpoint incidence 79 61 52 29 11 Control 78 69 67 48 13 Early surgery Participants, n 0 20 40 60 80 Time (Months) Early surgery Conservative treatment Figure 2 Cumulative incidence of primary composite outcome (intention-to-treat population) Table 3 Indications for aortic valve replacement in the conservative treatment group Symptom onset (including increased fatigue) 18 (51.4%) AS progression 6 (17.1%) Decrease in LVEF 3 (8.6%) Combination of factors 8 (22.9%) AS, aortic stenosis; LVEF, left ventricular ejection fraction. Aortic valve replacement vs. conservative treatment 4531 Downloaded from academic.oup.com/eurheartj/article/45/42/4526/7742145 by guest on 03 October 2026 [Extracted table 1] \| Symptom onset (including increased fatigue) 1\|8 (51.4%)\| \|---\|---\| \|AS progression 6 (17.1%)\|\| \|Decrease in LVEF\|3 (8.6%)\| \|Combination of factors 8 (22.9%)\|\| |
| eh_d54a2600ddc753b3 | ClinicalTrials.gov NCT02436655 | 1 | 95–101 | d32e23eb2a1e | ### Primary outcome 1: all cause death, Major Adverse Cardiac Event (MACE) including: ( Acute Myocardial Infarction - AMI, Stroke - CVI, unplanned hospitalization for Heart failure (HF) needing intravenous treatment - Measure: all cause death, Major Adverse Cardiac Event (MACE) including: ( Acute Myocardial Infarction - AMI, Stroke - CVI, unplanned hospitalization for Heart failure (HF) needing intravenous treatment - Time frame: 36 months ### Primary outcome 2: all cause death, Major Adverse Cardiac Event (MACE) including: ( Acute Myocardial Infarction - AMI, Stroke - CVI, unplanned hospitalization for Heart failure (HF) needing intravenous treatment - Measure: all cause death, Major Adverse Cardiac Event (MACE) including: ( Acute Myocardial Infarction - AMI, Stroke - CVI, unplanned hospitalization for Heart failure (HF) needing intravenous treatment - Time frame: 5 years |
| eh_d6c269341c7d8a5d | CIRCULATIONAHA.121.057639.pdf | 3 | 46–50 | 592b9a6c531c | randomly assigned to early surgery or conservative treatment using a web-based interactive response system. The assign- ment to each treatment group was computer-generated and stratified according to the participating centers by means of a permuted-block sequence with variable block size. Details |
| eh_e66d374e838e8c6d | ehae585.pdf | 6 | 26–32 | a890f771fd05 | nificant heterogeneity between both groups was noted. All patients were included in the analysis. Median follow-up for all patients was 63 months [interquartile range (IQR) 48-75). Median follow-up in sur vivors was 68 months (IQR 58-79). One patient, who was randomized to early surgery, has been lost to follow-up, and this patient was cen sored when the last follow-up information was available and included in the intention-to-treat analysis (Figure 1). |
| eh_f625637d366df2a0 | CIRCULATIONAHA.121.057639.pdf | 4 | 62–69 | 592b9a6c531c | were compared by using 2-sample t tests. A Kaplan-Meier estimator was used to estimate the distribution of time to pri- mary and time-to-event secondary end points (eg, survival and time to first HF hospitalization) and a log-rank test to compare them between the 2 treatment groups. A Cox proportional- hazards regression model that included treatment was used to estimate the hazard ratio comparing the early surgery and the conservative treatment groups. For dichotomous second- |

Illustrative all-cause-death assessment (Banovic 2024). Part A gives all questions, answers, evidence types, justifications, unknowns, counterevidence and Unassisted rationales. — means none recorded in those fields; inactive questions follow branching. Part B preserves 16 archived records (15 cited by current answers). Quotes and rationales are verbatim; quote line breaks display as spaces. Source

| Handle | Source | Page | Lines | Source-version hash | Full verbatim quote |
|---|---|---|---|---|---|

locations and hashes identify captured versions. References within quotations concern original documents. Full trace: analysis/data-derived/avatar-trace.json. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 2. Benchmark trials span sixteen meta-analyses**

| Specialty | Journal and year | Topic | Trials | Published D1–D5 labels (Low/SC/High) | Overall given |
|---|---|---|---|---|---|
| General medicine | JAMA 2026 | SGLT2 inhibitors and kidney outcomes across eGFR and albuminuria | 7 | 35 / 0 / 0 | No |
| General medicine | Lancet 2025 | Clopidogrel vs aspirin for secondary prevention of coronary artery disease | 7 | 32 / 3 / 0 | Yes |
| Cardiology | J Am Coll Cardiol 2025 | Early aortic valve replacement vs surveillance in asymptomatic severe aortic stenosis | 2 | 10 / 0 / 0 | Yes |
| Endocrinology | Lancet Diabetes Endocrinol 2025 | GLP-1 receptor agonists and kidney and cardiovascular outcomes | 7 | 35 / 0 / 0 | Yes |
| Gastroenterology | Gastroenterology 2025 | GLP-1 receptor agonists and gastrointestinal adverse events | 7 | 31 / 4 / 0 | Yes |
| Infectious diseases | Clin Infect Dis 2025 | Safety of high-dose rifampin in tuberculosis treatment | 6 | 30 / 0 / 0 | Yes |
| Nephrology | Clin J Am Soc Nephrol 2025 | Therapies for proteinuria and eGFR decline in IgA nephropathy | 7 | 30 / 5 / 0 | No |
| Neurology | Lancet Neurol 2025 | Pharmacotherapy and non-invasive neuromodulation for neuropathic pain | 7 | 24 / 7 / 4 | Yes |
| Obstetrics and gynaecology | Hum Reprod Update 2026 | Preconception lifestyle interventions for women | 7 | 23 / 10 / 2 | Yes |
| Oncology | Lancet Oncol 2026 | CDK4/6 inhibitors plus endocrine therapy in HR-positive, HER2-negative advanced breast cancer | 7 | 35 / 0 / 0 | Yes |
| Orthopaedics | Am J Sports Med 2025 | Lateral extra-articular procedure added to primary ACL reconstruction | 5 | 20 / 3 / 2 | Yes |
| Paediatrics | JAMA Pediatr 2025 | Active treatment vs expectant management of patent ductus arteriosus in preterm infants | 7 | 29 / 6 / 0 | Yes |
| Psychiatry | Lancet Psychiatry 2026 | Cannabinoids for mental and substance use disorders | 5 | 19 / 6 / 0 | Yes |
| Pulmonary and critical care | Lancet Respir Med 2025 | Preoxygenation strategies for intubation in critically ill patients | 7 | 32 / 0 / 3 | No |
| Rheumatology | Lancet Rheumatol 2025 | Long-term effectiveness of non-surgical interventions for chronic low back pain | 7 | 12 / 23 / 0 | Yes |
| Surgery and anaesthesia | Br J Anaesth 2025 | Home-based prehabilitation before surgery | 7 | 30 / 2 / 3 | Yes |
| **Total** | | | 102 | 427 / 69 / 14 | 81 / 102 |

102 trials; SC, Some concerns. Overall given indicates availability of published Overall labels. ACL, anterior cruciate ligament; CDK4/6, cyclin-dependent kinases 4 and 6; eGFR, estimated glomerular filtration rate; GLP-1, glucagon-like peptide-1; HER2, human epidermal growth factor receptor 2; HR, hormone receptor; IgA, immunoglobulin A; SGLT2, sodium-glucose cotransporter 2. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 3. Published label distributions by domain**

| Specialty | Topic | Trials | D1 | D2 | D3 | D4 | D5 |
|---|---|---|---|---|---|---|---|
| General medicine | Clopidogrel vs aspirin for secondary prevention of coronary artery disease | 7 | 7/0/0 | 4/3/0 | 7/0/0 | 7/0/0 | 7/0/0 |
| General medicine | SGLT2 inhibitors and kidney outcomes across eGFR and albuminuria | 7 | 7/0/0 | 7/0/0 | 7/0/0 | 7/0/0 | 7/0/0 |
| Cardiology | Early aortic valve replacement vs surveillance in asymptomatic severe aortic stenosis | 2 | 2/0/0 | 2/0/0 | 2/0/0 | 2/0/0 | 2/0/0 |
| Endocrinology | GLP-1 receptor agonists and kidney and cardiovascular outcomes | 7 | 7/0/0 | 7/0/0 | 7/0/0 | 7/0/0 | 7/0/0 |
| Gastroenterology | GLP-1 receptor agonists and gastrointestinal adverse events | 7 | 6/1/0 | 6/1/0 | 7/0/0 | 7/0/0 | 5/2/0 |
| Infectious diseases | Safety of high-dose rifampin in tuberculosis treatment | 6 | 6/0/0 | 6/0/0 | 6/0/0 | 6/0/0 | 6/0/0 |
| Nephrology | Therapies for proteinuria and eGFR decline in IgA nephropathy | 7 | 7/0/0 | 7/0/0 | 6/1/0 | 6/1/0 | 4/3/0 |
| Neurology | Pharmacotherapy and non-invasive neuromodulation for neuropathic pain | 7 | 6/1/0 | 6/1/0 | 3/3/1 | 7/0/0 | 2/2/3 |
| Obstetrics and gynaecology | Preconception lifestyle interventions for women | 7 | 3/3/1 | 7/0/0 | 5/1/1 | 6/1/0 | 2/5/0 |
| Oncology | CDK4/6 inhibitors plus endocrine therapy in HR-positive, HER2-negative advanced breast cancer | 7 | 7/0/0 | 7/0/0 | 7/0/0 | 7/0/0 | 7/0/0 |
| Orthopaedics | Lateral extra-articular procedure added to primary ACL reconstruction | 5 | 2/1/2 | 4/1/0 | 5/0/0 | 4/1/0 | 5/0/0 |
| Paediatrics | Active treatment vs expectant management of patent ductus arteriosus in preterm infants | 7 | 6/1/0 | 3/4/0 | 7/0/0 | 6/1/0 | 7/0/0 |
| Psychiatry | Cannabinoids for mental and substance use disorders | 5 | 4/1/0 | 3/2/0 | 3/2/0 | 4/1/0 | 5/0/0 |
| Pulmonary and critical care | Preoxygenation strategies for intubation in critically ill patients | 7 | 7/0/0 | 7/0/0 | 4/0/3 | 7/0/0 | 7/0/0 |
| Rheumatology | Long-term effectiveness of non-surgical interventions for chronic low back pain | 7 | 3/4/0 | 5/2/0 | 0/7/0 | 3/4/0 | 1/6/0 |
| Surgery and anaesthesia | Home-based prehabilitation before surgery | 7 | 4/1/2 | 5/1/1 | 7/0/0 | 7/0/0 | 7/0/0 |

Each D1–D5 cell gives Low / Some concerns / High counts; 102 trials. Source journals and years are in the preceding benchmark eTable. ACL, anterior cruciate ligament; CDK4/6, cyclin-dependent kinases 4 and 6; eGFR, estimated glomerular filtration rate; GLP-1, glucagon-like peptide-1; HER2, human epidermal growth factor receptor 2; HR, hormone receptor; IgA, immunoglobulin A; SGLT2, sodium-glucose cotransporter 2. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 4. Confusion matrices against Published**

**ARBITER (Sonnet 5.5) — D1**

| Published | Low | Some concerns | High |
|---|---|---|---|
| Low | 75 | 6 | 3 |
| Some concerns | 7 | 6 | 0 |
| High | 1 | 3 | 1 |

**ARBITER (Sonnet 5.5) — D2**

| Published | Low | Some concerns | High |
|---|---|---|---|
| Low | 69 | 14 | 3 |
| Some concerns | 12 | 2 | 1 |
| High | 1 | 0 | 0 |

**ARBITER (Sonnet 5.5) — D3**

| Published | Low | Some concerns | High |
|---|---|---|---|
| Low | 66 | 13 | 4 |
| Some concerns | 1 | 7 | 6 |
| High | 1 | 2 | 2 |

Each grid contains counts of cells: Published rows × predicted columns, ordered Low / Some concerns / High. Missing predictions appear only when nonzero. Overall uses the 81 labelled trials. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 4 (continued). Confusion matrices against Published**

**ARBITER (Sonnet 5.5) — D4**

| Published | Low | Some concerns | High |
|---|---|---|---|
| Low | 87 | 6 | 0 |
| Some concerns | 2 | 6 | 1 |
| High | 0 | 0 | 0 |

**ARBITER (Sonnet 5.5) — D5**

| Published | Low | Some concerns | High |
|---|---|---|---|
| Low | 50 | 31 | 0 |
| Some concerns | 3 | 15 | 0 |
| High | 0 | 3 | 0 |

**ARBITER (Sonnet 5.5) — Overall**

| Published | Low | Some concerns | High |
|---|---|---|---|
| Low | 23 | 15 | 4 |
| Some concerns | 3 | 15 | 4 |
| High | 0 | 8 | 9 |

Each grid contains counts of cells: Published rows × predicted columns, ordered Low / Some concerns / High. Missing predictions appear only when nonzero. Overall uses the 81 labelled trials. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 4 (continued). Confusion matrices against Published**

**Unassisted (Sonnet 5.5) — D1**

| Published | Low | Some concerns | High |
|---|---|---|---|
| Low | 75 | 7 | 2 |
| Some concerns | 6 | 7 | 0 |
| High | 1 | 4 | 0 |

**Unassisted (Sonnet 5.5) — D2**

| Published | Low | Some concerns | High |
|---|---|---|---|
| Low | 71 | 11 | 4 |
| Some concerns | 12 | 2 | 1 |
| High | 0 | 1 | 0 |

**Unassisted (Sonnet 5.5) — D3**

| Published | Low | Some concerns | High |
|---|---|---|---|
| Low | 70 | 7 | 6 |
| Some concerns | 4 | 6 | 4 |
| High | 4 | 0 | 1 |

Each grid contains counts of cells: Published rows × predicted columns, ordered Low / Some concerns / High. Missing predictions appear only when nonzero. Overall uses the 81 labelled trials. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 4 (continued). Confusion matrices against Published**

**Unassisted (Sonnet 5.5) — D4**

| Published | Low | Some concerns | High |
|---|---|---|---|
| Low | 89 | 4 | 0 |
| Some concerns | 4 | 2 | 3 |
| High | 0 | 0 | 0 |

**Unassisted (Sonnet 5.5) — D5**

| Published | Low | Some concerns | High |
|---|---|---|---|
| Low | 61 | 19 | 1 |
| Some concerns | 8 | 10 | 0 |
| High | 2 | 0 | 1 |

**Unassisted (Sonnet 5.5) — Overall**

| Published | Low | Some concerns | High |
|---|---|---|---|
| Low | 32 | 7 | 3 |
| Some concerns | 12 | 7 | 3 |
| High | 2 | 6 | 9 |

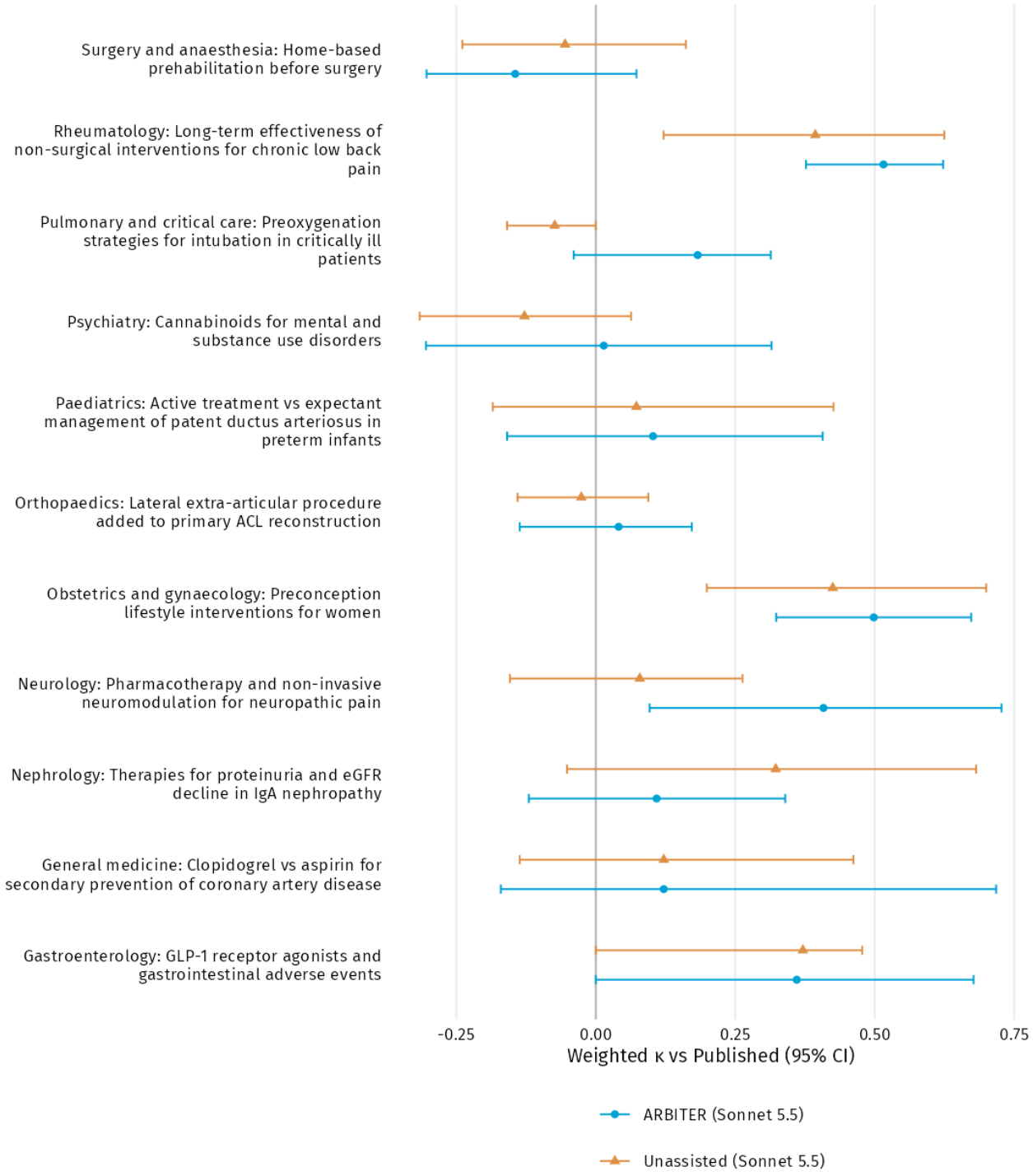
Each grid contains counts of cells: Published rows × predicted columns, ordered Low / Some concerns / High. Missing predictions appear only when nonzero. Overall uses the 81 labelled trials. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 5. Per-meta-analysis weighted κ**

| Specialty | Topic | ARBITER (Sonnet 5.5) | Unassisted (Sonnet 5.5) |
|---|---|---|---|
| General medicine | Clopidogrel vs aspirin for secondary prevention of coronary artery disease | 0.12 (−0.17 to 0.72) | 0.12 (−0.14 to 0.46) |
| General medicine | SGLT2 inhibitors and kidney outcomes across eGFR and albuminuria | — | — |
| Cardiology | Early aortic valve replacement vs surveillance in asymptomatic severe aortic stenosis | — | — |
| Endocrinology | GLP-1 receptor agonists and kidney and cardiovascular outcomes | — | — |
| Gastroenterology | GLP-1 receptor agonists and gastrointestinal adverse events | 0.36 (0.00–0.68) | 0.37 (0.00–0.48) |
| Infectious diseases | Safety of high-dose rifampin in tuberculosis treatment | — | — |
| Nephrology | Therapies for proteinuria and eGFR decline in IgA nephropathy | 0.11 (−0.12 to 0.34) | 0.32 (−0.05 to 0.68) |
| Neurology | Pharmacotherapy and non-invasive neuromodulation for neuropathic pain | 0.41 (0.10–0.73) | 0.08 (−0.15 to 0.26) |
| Obstetrics and gynaecology | Preconception lifestyle interventions for women | 0.50 (0.32–0.67) | 0.42 (0.20–0.70) |
| Oncology | CDK4/6 inhibitors plus endocrine therapy in HR-positive, HER2-negative advanced breast cancer | — | — |
| Orthopaedics | Lateral extra-articular procedure added to primary ACL reconstruction | 0.04 (−0.14 to 0.17) | −0.03 (−0.14 to 0.09) |
| Paediatrics | Active treatment vs expectant management of patent ductus arteriosus in preterm infants | 0.10 (−0.16 to 0.41) | 0.07 (−0.18 to 0.43) |
| Psychiatry | Cannabinoids for mental and substance use disorders | 0.01 (−0.30 to 0.32) | −0.13 (−0.32 to 0.06) |
| Pulmonary and critical care | Preoxygenation strategies for intubation in critically ill patients | 0.18 (−0.04 to 0.31) | −0.07 (−0.16 to 0.00) |
| Rheumatology | Long-term effectiveness of non-surgical interventions for chronic low back pain | 0.52 (0.38–0.62) | 0.39 (0.12–0.62) |
| Surgery and anaesthesia | Home-based prehabilitation before surgery | −0.14 (−0.30 to 0.07) | −0.06 (−0.24 to 0.16) |

ACL, anterior cruciate ligament; CDK4/6, cyclin-dependent kinases 4 and 6; eGFR, estimated glomerular filtration rate; GLP-1, glucagon-like peptide-1; HER2, human epidermal growth factor receptor 2; HR, hormone receptor; IgA, immunoglobulin A; SGLT2, sodium-glucose cotransporter 2. Weighted κ (95% CI). eFigure 1 displays the defined nondegenerate estimates. A coefficient with no variance, or a degenerate bootstrap interval, is shown as —. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eFigure 1. Agreement with Published varied across meta-analyses**




Weighted kappa and 95% CIs; source symbols distinguish model and assistance. Undefined coefficients are omitted. Specialty and topic identify each meta-analysis. Descriptive estimates only; no statistical tests. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018.

**eTable 6. Agreement against Published, all sources and domains**

**ARBITER (Sonnet 5.5)**

| Domain | Weighted κ | AC2 | Exact agreement | Sensitivity | Specificity |
|---|---|---|---|---|---|
| D1–D5 | 0.34 (0.26–0.42) | 0.81 (0.78–0.84) | 386/510 (75.7%; 72.4–79.0) | 55/83 (66.3%; 57.3–74.7) | 347/427 (81.3%; 77.7–84.7) |
| D1 | 0.38 (0.19–0.55) | 0.84 (0.78–0.89) | 82/102 (80.4%; 74.5–86.3) | 10/18 (55.6%; 35.0–77.8) | 75/84 (89.3%; 83.2–95.1) |
| D2 | −0.01 (−0.15 to 0.14) | 0.77 (0.69–0.84) | 71/102 (69.6%; 61.8–77.5) | 3/16 (18.8%; 0.0–37.5) | 69/86 (80.2%; 72.6–87.4) |
| D3 | 0.43 (0.29–0.57) | 0.76 (0.68–0.84) | 75/102 (73.5%; 66.7–80.4) | 17/19 (89.5%; 73.9–100.0) | 66/83 (79.5%; 72.6–86.4) |
| D4 | 0.57 (0.33–0.78) | 0.95 (0.91–0.98) | 93/102 (91.2%; 86.3–96.1) | 7/9 (77.8%; 50.0–100.0) | 87/93 (93.5%; 88.9–97.8) |
| D5 | 0.30 (0.20–0.41) | 0.71 (0.64–0.77) | 65/102 (63.7%; 56.9–71.6) | 18/21 (85.7%; 75.0–95.5) | 50/81 (61.7%; 53.1–70.4) |
| Overall | 0.44 (0.32–0.56) | 0.50 (0.39–0.60) | 47/81 (58.0%; 49.4–66.7) | 36/39 (92.3%; 83.7–100.0) | 23/42 (54.8%; 44.2–67.4) |

Linearly weighted Cohen's κ and Gwet's AC2. Exact agreement, sensitivity and specificity are n/N (%); their CI bounds are percentages. Some concerns or High is the positive class. All-Low rule has no variance; only accuracy and specificity are reported. Paired differences are on the coefficient scale or in percentage points; — indicates undefined or not applicable. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 6 (continued). Agreement against Published, all sources and domains**

**Unassisted (Sonnet 5.5)**

| Domain | Weighted κ | AC2 | Exact agreement | Sensitivity | Specificity |
|---|---|---|---|---|---|
| D1–D5 | 0.28 (0.21–0.35) | 0.82 (0.79–0.85) | 395/510 (77.5%; 74.1–80.8) | 42/83 (50.6%; 40.3–60.2) | 366/427 (85.7%; 82.1–89.3) |
| D1 | 0.39 (0.21–0.55) | 0.85 (0.79–0.90) | 82/102 (80.4%; 74.5–86.3) | 11/18 (61.1%; 40.0–81.8) | 75/84 (89.3%; 83.3–95.1) |
| D2 | 0.06 (−0.10 to 0.21) | 0.79 (0.72–0.85) | 73/102 (71.6%; 64.7–78.4) | 4/16 (25.0%; 6.7–45.5) | 71/86 (82.6%; 76.1–89.0) |
| D3 | 0.29 (0.13–0.44) | 0.76 (0.67–0.83) | 77/102 (75.5%; 68.6–82.4) | 11/19 (57.9%; 38.9–76.5) | 70/83 (84.3%; 77.9–90.7) |
| D4 | 0.43 (0.22–0.61) | 0.94 (0.90–0.97) | 91/102 (89.2%; 83.3–94.1) | 5/9 (55.6%; 28.6–85.7) | 89/93 (95.7%; 92.0–98.9) |
| D5 | 0.25 (0.07–0.43) | 0.76 (0.69–0.83) | 72/102 (70.6%; 63.7–78.4) | 11/21 (52.4%; 34.8–71.4) | 61/81 (75.3%; 67.9–82.9) |
| Overall | 0.42 (0.27–0.56) | 0.53 (0.43–0.64) | 48/81 (59.3%; 50.6–67.9) | 25/39 (64.1%; 50.0–77.8) | 32/42 (76.2%; 65.9–86.8) |

Linearly weighted Cohen's κ and Gwet's AC2. Exact agreement, sensitivity and specificity are n/N (%); their CI bounds are percentages. Some concerns or High is the positive class. All-Low rule has no variance; only accuracy and specificity are reported. Paired differences are on the coefficient scale or in percentage points; — indicates undefined or not applicable. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 6 (continued). Agreement against Published, all sources and domains**

**All-Low rule**

| Domain | Weighted κ | AC2 | Exact agreement | Sensitivity | Specificity |
|---|---|---|---|---|---|
| D1–D5 | — | — | 427/510 (83.7%; 81.0–86.5) | — | 427/427 (100.0%) |
| D1 | — | — | 84/102 (82.4%; 76.5–88.2) | — | 84/84 (100.0%) |
| D2 | — | — | 86/102 (84.3%; 78.4–90.2) | — | 86/86 (100.0%) |
| D3 | — | — | 83/102 (81.4%; 76.5–86.3) | — | 83/83 (100.0%) |
| D4 | — | — | 93/102 (91.2%; 86.3–96.1) | — | 93/93 (100.0%) |
| D5 | — | — | 81/102 (79.4%; 74.5–84.3) | — | 81/81 (100.0%) |
| Overall | — | — | 42/81 (51.9%; 44.4–59.3) | — | 42/42 (100.0%) |

Linearly weighted Cohen's κ and Gwet's AC2. Exact agreement, sensitivity and specificity are n/N (%); their CI bounds are percentages. Some concerns or High is the positive class. All-Low rule has no variance; only accuracy and specificity are reported. Paired differences are on the coefficient scale or in percentage points; — indicates undefined or not applicable. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 7. Paired differences against Published**

| Domain | Weighted κ | AC2 | Exact agreement | Sensitivity | Specificity |
|---|---|---|---|---|---|
| **ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)** | | | | | |
| D1–D5 | 0.06 (−0.03 to 0.14) | −0.01 (−0.04 to 0.02) | −1.8 (−5.1 to 1.6) | 15.7 (6.8–24.4) | −4.4 (−7.6 to −1.2) |
| D1 | −0.01 (−0.12 to 0.10) | −0.01 (−0.04 to 0.03) | 0.0 (−3.9 to 3.9) | −5.6 (−16.7 to 0.0) | 0.0 (−2.6 to 2.6) |
| D2 | −0.07 (−0.24 to 0.11) | −0.01 (−0.09 to 0.06) | −2.0 (−9.8 to 5.9) | −6.2 (−26.7 to 13.3) | −2.3 (−11.5 to 6.6) |
| D3 | 0.14 (0.01–0.28) | 0.00 (−0.05 to 0.06) | −2.0 (−7.8 to 3.9) | 31.6 (13.6–47.6) | −4.8 (−10.5 to 0.0) |
| D4 | 0.13 (−0.08 to 0.36) | 0.01 (−0.02 to 0.05) | 2.0 (−3.9 to 7.8) | 22.2 (0.0–50.0) | −2.2 (−6.6 to 2.2) |
| D5 | 0.05 (−0.15 to 0.24) | −0.05 (−0.13 to 0.02) | −6.9 (−15.7 to 2.0) | 33.3 (11.8–52.9) | −13.6 (−22.0 to −5.0) |
| Overall | 0.02 (−0.11 to 0.16) | −0.04 (−0.16 to 0.09) | −1.2 (−11.1 to 8.6) | 28.2 (16.1–40.5) | −21.4 (−32.7 to −9.8) |

ARBITER minus Unassisted; coefficients in original units, proportions in percentage points. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 8. Post hoc: unweighted agreement with published labels is modest**

| Source | Domain | Measure | Estimate (95% CI) |
|---|---|---|---|
| ARBITER (Sonnet 5.5) | D1 | Unweighted κ | 0.37 (0.18–0.53) |
| ARBITER (Sonnet 5.5) | D1 | Exact agreement | 82/102 (80.4%; 74.5–86.3) |
| ARBITER (Sonnet 5.5) | D2 | Unweighted κ | −0.02 (−0.16 to 0.14) |
| ARBITER (Sonnet 5.5) | D2 | Exact agreement | 71/102 (69.6%; 61.8–77.5) |
| ARBITER (Sonnet 5.5) | D3 | Unweighted κ | 0.37 (0.23–0.51) |
| ARBITER (Sonnet 5.5) | D3 | Exact agreement | 75/102 (73.5%; 66.7–80.4) |
| ARBITER (Sonnet 5.5) | D4 | Unweighted κ | 0.55 (0.30–0.77) |
| ARBITER (Sonnet 5.5) | D4 | Exact agreement | 93/102 (91.2%; 86.3–96.1) |
| ARBITER (Sonnet 5.5) | D5 | Unweighted κ | 0.28 (0.18–0.39) |
| ARBITER (Sonnet 5.5) | D5 | Exact agreement | 65/102 (63.7%; 56.9–71.6) |
| ARBITER (Sonnet 5.5) | Overall | Unweighted κ | 0.37 (0.24–0.49) |
| ARBITER (Sonnet 5.5) | Overall | Exact agreement | 47/81 (58.0%; 49.4–66.7) |
| Unassisted (Sonnet 5.5) | D1 | Unweighted κ | 0.38 (0.20–0.54) |
| Unassisted (Sonnet 5.5) | D1 | Exact agreement | 82/102 (80.4%; 74.5–86.3) |
| Unassisted (Sonnet 5.5) | D2 | Unweighted κ | 0.03 (−0.11 to 0.19) |
| Unassisted (Sonnet 5.5) | D2 | Exact agreement | 73/102 (71.6%; 64.7–78.4) |
| Unassisted (Sonnet 5.5) | D3 | Unweighted κ | 0.31 (0.15–0.46) |
| Unassisted (Sonnet 5.5) | D3 | Exact agreement | 77/102 (75.5%; 68.6–82.4) |
| Unassisted (Sonnet 5.5) | D4 | Unweighted κ | 0.34 (0.13–0.55) |
| Unassisted (Sonnet 5.5) | D4 | Exact agreement | 91/102 (89.2%; 83.3–94.1) |
| Unassisted (Sonnet 5.5) | D5 | Unweighted κ | 0.26 (0.09–0.42) |
| Unassisted (Sonnet 5.5) | D5 | Exact agreement | 72/102 (70.6%; 63.7–78.4) |
| Unassisted (Sonnet 5.5) | Overall | Unweighted κ | 0.32 (0.17–0.47) |
| Unassisted (Sonnet 5.5) | Overall | Exact agreement | 48/81 (59.3%; 50.6–67.9) |

Post hoc. Unweighted Cohen's κ and exact agreement against published review labels, by domain and Overall. κ here is unweighted; exact agreement uses labelled trials. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 9. Adjudicators usually agreed with models when they disagreed with Published**

| Source | Direction | Outcome | Count | Consensus n | All n |
|---|---|---|---|---|---|
| ARBITER | More lenient | Other label | 0 | 28 | 36 |
| ARBITER | More lenient | Published upheld | 6 | 28 | 36 |
| ARBITER | More lenient | Split | 8 | 28 | 36 |
| ARBITER | More lenient | Adjudicators agreed with it | 22 | 28 | 36 |
| ARBITER | Stricter | Other label | 2 | 77 | 88 |
| ARBITER | Stricter | Published upheld | 2 | 77 | 88 |
| ARBITER | Stricter | Split | 11 | 77 | 88 |
| ARBITER | Stricter | Adjudicators agreed with it | 73 | 77 | 88 |
| Unassisted | More lenient | Other label | 2 | 38 | 46 |
| Unassisted | More lenient | Published upheld | 17 | 38 | 46 |
| Unassisted | More lenient | Split | 8 | 38 | 46 |
| Unassisted | More lenient | Adjudicators agreed with it | 19 | 38 | 46 |
| Unassisted | Stricter | Other label | 1 | 59 | 69 |
| Unassisted | Stricter | Published upheld | 10 | 59 | 69 |
| Unassisted | Stricter | Split | 10 | 59 | 69 |
| Unassisted | Stricter | Adjudicators agreed with it | 48 | 59 | 69 |

Both sources use the same model. Figure 4B uses disagreements where both adjudicators agreed as the denominator; cells where adjudicators split are displayed separately. Adjudicators agreed with it means their agreed choice equals the source label; Published upheld means their agreed choice equals the published review label; Other label means neither. Count is the outcome count; Agreed n excludes divided decisions and All n includes them. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 10. Published labels were usually less severe than the agreed choice**

| Domain | Direction | Eligible cells (n) | Total |
|---|---|---|---|
| D1 | More lenient | 9 | 18 |
| D1 | Equal | 2 | 18 |
| D1 | Stricter | 7 | 18 |
| D2 | More lenient | 11 | 32 |
| D2 | Equal | 11 | 32 |
| D2 | Stricter | 10 | 32 |
| D3 | More lenient | 19 | 24 |
| D3 | Equal | 5 | 24 |
| D3 | Stricter | 0 | 24 |
| D4 | More lenient | 9 | 14 |
| D4 | Equal | 3 | 14 |
| D4 | Stricter | 2 | 14 |
| D5 | More lenient | 30 | 41 |
| D5 | Equal | 8 | 41 |
| D5 | Stricter | 3 | 41 |
| All | More lenient | 78 | 129 |
| All | Equal | 29 | 129 |
| All | Stricter | 22 | 129 |

Published review label relative to the adjudicators' agreed choice, on the 129 ratings where both adjudicators agreed. Directions compare severity on Low < Some concerns < High. Selected disputed ratings cannot estimate full-benchmark accuracy. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 11. Preferred shares among adjudicated arm disagreements: sensitivities**

| Analysis | Measure | Estimate (95% CI) |
|---|---|---|
| Both adjudicators agreed | ARBITER | 54/62 (87.1%; 79.0–94.3) |
| Both adjudicators agreed | Unassisted | 7/62 (11.3%; 4.5–19.2) |
| Both adjudicators agreed | Neither | 1/62 (1.6%; 0.0–4.8) |
| Adjudicator 1 | ARBITER | 60/80 (75.0%; 66.7–83.3) |
| Adjudicator 1 | Unassisted | 16/80 (20.0%; 11.7–28.9) |
| Adjudicator 1 | Neither | 4/80 (5.0%; 1.3–9.2) |
| Adjudicator 2 | ARBITER | 64/80 (80.0%; 72.3–87.2) |
| Adjudicator 2 | Unassisted | 15/80 (18.8%; 11.6–26.3) |
| Adjudicator 2 | Neither | 1/80 (1.2%; 0.0–3.8) |
| Counting split decisions against ARBITER | ARBITER | 54/80 (67.5%; 58.7–76.3) |
| Counting split decisions against ARBITER | Unassisted | 7/80 (8.8%; 3.5–15.0) |
| Counting split decisions against ARBITER | Neither | 19/80 (23.8%; 16.0–31.3) |

Fractions are adjudicated arm disagreements where the named rating was preferred / all adjudicated arm disagreements in that analysis. Consensus is primary and includes only cells with the same choice from both adjudicators; each single-adjudicator analysis includes cells where adjudicators split. Split cells are forced ARBITER losses in the split sensitivity; Neither includes these forced losses. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 12. Post hoc: preferred shares among adjudicated arm disagreements by strictness**

| Direction | Measure | Estimate (95% CI) |
|---|---|---|
| ARBITER stricter than Unassisted | ARBITER | 44/45 (97.8%; 92.7–100.0) |
| ARBITER stricter than Unassisted | Unassisted | 1/45 (2.2%; 0.0–7.3) |
| ARBITER stricter than Unassisted | Neither | 0/45 (0.0%) |
| ARBITER more lenient than Unassisted | ARBITER | 10/17 (58.8%; 35.7–80.0) |
| ARBITER more lenient than Unassisted | Unassisted | 6/17 (35.3%; 14.3–57.9) |
| ARBITER more lenient than Unassisted | Neither | 1/17 (5.9%; 0.0–18.2) |

Post hoc analysis among adjudicated arm disagreements where both adjudicators agreed, stratified by ARBITER severity relative to Unassisted. Fractions are cells where the named rating was preferred / cells in that stratum. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 13. Independent and selected ratings: full agreement including AC2**

| Comparison | Weighted κ | AC2 | Exact agreement |
|---|---|---|---|
| Adjudicator 1 vs Adjudicator 2 independent | 0.59 (0.48–0.68) | 0.69 (0.61–0.77) | 114/154 (74.0%; 68.2–79.9) |
| Adjudicator 1 vs Adjudicator 2 preferred | 0.69 (0.57–0.79) | 0.83 (0.77–0.89) | 129/154 (83.8%; 78.6–88.9) |
| Adjudicator 1 independent vs published label | −0.08 (−0.17 to 0.01) | −0.06 (−0.18 to 0.07) | 34/154 (22.1%; 15.8–28.5) |
| Adjudicator 1 independent vs ARBITER (Sonnet 5.5) | 0.52 (0.41–0.62) | 0.66 (0.57–0.74) | 107/154 (69.5%; 62.7–76.0) |
| Adjudicator 1 independent vs Unassisted (Sonnet 5.5) | 0.22 (0.13–0.30) | 0.28 (0.18–0.38) | 69/154 (44.8%; 38.1–51.6) |
| Adjudicator 2 independent vs published label | −0.08 (−0.18 to 0.03) | 0.14 (0.04–0.24) | 42/154 (27.3%; 20.4–34.3) |
| Adjudicator 2 independent vs ARBITER (Sonnet 5.5) | 0.67 (0.56–0.76) | 0.79 (0.73–0.85) | 120/154 (77.9%; 71.7–84.0) |
| Adjudicator 2 independent vs Unassisted (Sonnet 5.5) | 0.31 (0.19–0.42) | 0.45 (0.37–0.53) | 76/154 (49.4%; 42.2–56.5) |

154 disputed cells per comparison, selected because at least one Sonnet arm differed from the published review label. Independent ratings preceded label presentation; selected ratings chose among three offered labels. κ and AC2 are linearly weighted; raw agreement is n/N (x.x%; 95% CI a–b). This selected population cannot estimate full-benchmark agreement. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 14. Localisation population eligibility**

| Source | Ratings not chosen (cells) | Primary eligible (cells) | Sensitivity |
|---|---|---|---|
| ARBITER (Sonnet 5.5) | 13 | 10 | 12 |
| Unassisted (Sonnet 5.5) | 60 | 47 | 57 |
| Published | 100 | — | — |

Primary requires both independent judgements equal to the adjudicators' agreed choice; sensitivity requires at least one. Published losses lack signalling answers. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 15. Error-localisation classes, both populations**

| Population | Class | ARBITER (Sonnet 5.5) | Unassisted (Sonnet 5.5) |
|---|---|---|---|
| Primary | Judgement departs from own answers | 0 | 4 |
| Primary | Single question that alone changed the rating | 6 | 27 |
| Primary | Several questions that alone changed the rating | 1 | 2 |
| Primary | Jointly decisive | 2 | 11 |
| Primary | Answers agree, judgement differs | 1 | 2 |
| Primary | Not attributable | 0 | 1 |
| Sensitivity | Judgement departs from own answers | 0 | 4 |
| Sensitivity | Single question that alone changed the rating | 6 | 29 |
| Sensitivity | Several questions that alone changed the rating | 1 | 2 |
| Sensitivity | Jointly decisive | 4 | 18 |
| Sensitivity | Answers agree, judgement differs | 1 | 2 |
| Sensitivity | Not attributable | 0 | 2 |

Counts of losing cells. First applicable class is assigned; classes sum to eligible losses. No-discordance can include adjudicator disagreements, flagged separately. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 16. Questions that alone changed the rating: frequencies, both populations**

| Population | Domain | Question | Signalling question | ARBITER (Sonnet 5.5) | Unassisted (Sonnet 5.5) |
|---|---|---|---|---|---|
| Primary | D1 | 1.1 | Random allocation sequence | 0 | 0 |
| Primary | D1 | 1.2 | Concealment until enrolment and assignment | 1 | 3 |
| Primary | D1 | 1.3 | Preintervention imbalance suggesting a randomisation problem | 0 | 0 |
| Primary | D2 | 2.1 | Participants aware of assignment | 0 | 0 |
| Primary | D2 | 2.2 | Carers or intervention providers aware of assignment | 0 | 0 |
| Primary | D2 | 2.3 | Deviations arising from trial context | 3 | 3 |
| Primary | D2 | 2.4 | Deviations affecting the outcome | 0 | 0 |
| Primary | D2 | 2.5 | Deviations balanced across groups | 0 | 0 |
| Primary | D2 | 2.6 | Appropriate analysis of assignment effect | 0 | 3 |
| Primary | D2 | 2.7 | Substantial impact of analysing outside assigned groups | 1 | 0 |
| Primary | D3 | 3.1 | Outcome data available for nearly everyone | 0 | 1 |
| Primary | D3 | 3.2 | Evidence against bias from missing data | 0 | 0 |
| Primary | D3 | 3.3 | Missingness potentially depends on true outcome | 0 | 0 |
| Primary | D3 | 3.4 | Missingness likely depends on true outcome | 0 | 0 |
| Primary | D4 | 4.1 | Inappropriate measurement method | 0 | 0 |
| Primary | D4 | 4.2 | Measurement differs across groups | 0 | 0 |
| Primary | D4 | 4.3 | Assessors aware of received intervention | 0 | 0 |
| Primary | D4 | 4.4 | Assessment potentially influenced by awareness | 0 | 0 |
| Primary | D4 | 4.5 | Assessment likely influenced by awareness | 0 | 1 |
| Primary | D5 | 5.1 | Analysis plan finalised before unblinding | 2 | 18 |
| Primary | D5 | 5.2 | Result selected from eligible outcome measurements | 0 | 1 |
| Primary | D5 | 5.3 | Result selected from eligible analyses | 1 | 1 |
| Sensitivity | D1 | 1.1 | Random allocation sequence | 0 | 0 |
| Sensitivity | D1 | 1.2 | Concealment until enrolment and assignment | 1 | 3 |
| Sensitivity | D1 | 1.3 | Preintervention imbalance suggesting a randomisation problem | 0 | 0 |
| Sensitivity | D2 | 2.1 | Participants aware of assignment | 0 | 0 |

| Population | Domain | Question | Signalling question | ARBITER (Sonnet 5.5) | Unassisted (Sonnet 5.5) |
|---|---|---|---|---|---|
| Sensitivity | D2 | 2.2 | Carers or intervention providers aware of assignment | 0 | 0 |
| Sensitivity | D2 | 2.3 | Deviations arising from trial context | 3 | 3 |
| Sensitivity | D2 | 2.4 | Deviations affecting the outcome | 0 | 0 |
| Sensitivity | D2 | 2.5 | Deviations balanced across groups | 0 | 0 |
| Sensitivity | D2 | 2.6 | Appropriate analysis of assignment effect | 0 | 4 |
| Sensitivity | D2 | 2.7 | Substantial impact of analysing outside assigned groups | 1 | 0 |
| Sensitivity | D3 | 3.1 | Outcome data available for nearly everyone | 0 | 1 |
| Sensitivity | D3 | 3.2 | Evidence against bias from missing data | 0 | 0 |
| Sensitivity | D3 | 3.3 | Missingness potentially depends on true outcome | 0 | 0 |
| Sensitivity | D3 | 3.4 | Missingness likely depends on true outcome | 0 | 0 |
| Sensitivity | D4 | 4.1 | Inappropriate measurement method | 0 | 0 |
| Sensitivity | D4 | 4.2 | Measurement differs across groups | 0 | 0 |
| Sensitivity | D4 | 4.3 | Assessors aware of received intervention | 0 | 0 |
| Sensitivity | D4 | 4.4 | Assessment potentially influenced by awareness | 0 | 0 |
| Sensitivity | D4 | 4.5 | Assessment likely influenced by awareness | 0 | 1 |
| Sensitivity | D5 | 5.1 | Analysis plan finalised before unblinding | 2 | 18 |
| Sensitivity | D5 | 5.2 | Result selected from eligible outcome measurements | 0 | 2 |
| Sensitivity | D5 | 5.3 | Result selected from eligible analyses | 1 | 1 |

Counts of questions that alone changed the rating; all 22 questions are retained. A cell may contribute several. Jointly decisive sets contribute none. Short question texts follow the 2019 guidance; D2 uses effect of assignment. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 17. Unresolved independent answer**

| Trial | Domain | Adjudicator | Question | Recorded answer | Parsing issue |
|---|---|---|---|---|---|
| Chiba 2021 | D3 | Adjudicator 1 | 3.4 | 3.4=NI at 6 months; PY at 12 months | Ambiguous or unrecognised answer |

The qualified answer is preserved without selecting a time point. This cell is outside both localisation populations because neither independent judgement equals the adjudicators' agreed choice. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 18. Exploratory signalling-answer agreement**

**Adjudicator 1**

| SQ | Signalling question | ARBITER (Sonnet 5.5) | Unassisted (Sonnet 5.5) |
|---|---|---|---|
| 1.1 | Random allocation sequence | 20/22 (90.9%; 78.3–100.0) | 20/21 (95.2%; 85.7–100.0) |
| 1.2 | Concealment until enrolment and assignment | 19/22 (86.4%; 72.0–100.0) | 16/21 (76.2%; 57.9–93.8) |
| 1.3 | Preintervention imbalance suggesting a randomisation problem | 12/22 (54.5%; 36.4–70.8) | 9/21 (42.9%; 23.5–60.9) |
| 2.1 | Participants aware of assignment | 34/40 (85.0%; 74.4–94.6) | 34/40 (85.0%; 74.4–94.6) |
| 2.2 | Carers or intervention providers aware of assignment | 37/40 (92.5%; 84.2–100.0) | 36/40 (90.0%; 81.0–97.6) |
| 2.3 | Deviations arising from trial context | 27/40 (67.5%; 53.1–80.6) | 27/40 (67.5%; 53.8–81.0) |
| 2.4 | Deviations affecting the outcome | 39/40 (97.5%; 92.5–100.0) | 38/40 (95.0%; 87.8–100.0) |
| 2.5 | Deviations balanced across groups | 39/40 (97.5%; 92.5–100.0) | 39/40 (97.5%; 92.5–100.0) |
| 2.6 | Appropriate analysis of assignment effect | 31/40 (77.5%; 65.7–88.6) | 23/40 (57.5%; 44.7–70.6) |
| 2.7 | Substantial impact of analysing outside assigned groups | 26/40 (65.0%; 51.4–77.5) | 17/40 (42.5%; 30.0–55.6) |
| 3.1 | Outcome data available for nearly everyone | 28/32 (87.5%; 75.7–97.0) | 19/31 (61.3%; 45.2–77.4) |
| 3.2 | Evidence against bias from missing data | 29/32 (90.6%; 80.0–100.0) | 19/32 (59.4%; 43.2–75.0) |
| 3.3 | Missingness potentially depends on true outcome | 28/32 (87.5%; 75.7–97.0) | 14/31 (45.2%; 29.2–61.5) |
| 3.4 | Missingness likely depends on true outcome | 16/32 (50.0%; 34.5–65.8) | 11/31 (35.5%; 20.7–51.5) |
| 4.1 | Inappropriate measurement method | 15/15 (100.0%) | 15/15 (100.0%) |
| 4.2 | Measurement differs across groups | 15/15 (100.0%) | 15/15 (100.0%) |
| 4.3 | Assessors aware of received intervention | 12/15 (80.0%; 58.8–100.0) | 13/15 (86.7%; 69.2–100.0) |
| 4.4 | Assessment potentially influenced by awareness | 13/15 (86.7%; 68.8–100.0) | 8/15 (53.3%; 31.2–75.0) |
| 4.5 | Assessment likely influenced by awareness | 12/15 (80.0%; 58.3–100.0) | 5/15 (33.3%; 11.1–56.2) |
| 5.1 | Analysis plan finalised before unblinding | 37/45 (82.2%; 73.8–90.2) | 19/45 (42.2%; 29.8–55.0) |
| 5.2 | Result selected from eligible outcome measurements | 21/45 (46.7%; 34.1–58.3) | 15/44 (34.1%; 22.5–45.2) |
| 5.3 | Result selected from eligible analyses | 6/45 (13.3%; 4.8–22.5) | 3/45 (6.7%; 0.0–13.6) |

SQ, signalling question. Correct collapsed answers / eligible questions (95% CI) on disputed domains. Independent answers define reference; Y/PY, N/PN, NI/NA collapse. Missing predictions are wrong. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 18 (continued). Exploratory signalling-answer agreement**

**Adjudicator 2**

| SQ | Signalling question | ARBITER (Sonnet 5.5) | Unassisted (Sonnet 5.5) |
|---|---|---|---|
| 1.1 | Random allocation sequence | 21/22 (95.5%; 85.7–100.0) | 19/21 (90.5%; 78.3–100.0) |
| 1.2 | Concealment until enrolment and assignment | 20/22 (90.9%; 78.9–100.0) | 14/21 (66.7%; 47.4–85.7) |
| 1.3 | Preintervention imbalance suggesting a randomisation problem | 22/22 (100.0%) | 19/21 (90.5%; 78.3–100.0) |
| 2.1 | Participants aware of assignment | 37/40 (92.5%; 84.4–100.0) | 37/40 (92.5%; 84.4–100.0) |
| 2.2 | Carers or intervention providers aware of assignment | 38/40 (95.0%; 88.2–100.0) | 37/40 (92.5%; 84.4–100.0) |
| 2.3 | Deviations arising from trial context | 31/40 (77.5%; 65.0–88.6) | 27/40 (67.5%; 54.3–80.0) |
| 2.4 | Deviations affecting the outcome | 39/40 (97.5%; 92.1–100.0) | 36/40 (90.0%; 80.6–97.6) |
| 2.5 | Deviations balanced across groups | 39/40 (97.5%; 92.1–100.0) | 39/40 (97.5%; 92.1–100.0) |
| 2.6 | Appropriate analysis of assignment effect | 32/40 (80.0%; 68.6–90.9) | 26/40 (65.0%; 51.4–77.5) |
| 2.7 | Substantial impact of analysing outside assigned groups | 31/40 (77.5%; 65.8–89.2) | 26/40 (65.0%; 51.7–77.8) |
| 3.1 | Outcome data available for nearly everyone | 30/32 (93.8%; 84.8–100.0) | 21/31 (67.7%; 52.8–82.8) |
| 3.2 | Evidence against bias from missing data | 30/32 (93.8%; 84.8–100.0) | 20/32 (62.5%; 46.9–78.1) |
| 3.3 | Missingness potentially depends on true outcome | 30/32 (93.8%; 84.8–100.0) | 16/31 (51.6%; 36.0–67.9) |
| 3.4 | Missingness likely depends on true outcome | 24/32 (75.0%; 60.6–88.0) | 11/31 (35.5%; 19.4–52.8) |
| 4.1 | Inappropriate measurement method | 15/15 (100.0%) | 15/15 (100.0%) |
| 4.2 | Measurement differs across groups | 15/15 (100.0%) | 15/15 (100.0%) |
| 4.3 | Assessors aware of received intervention | 12/15 (80.0%; 58.8–100.0) | 13/15 (86.7%; 69.2–100.0) |
| 4.4 | Assessment potentially influenced by awareness | 13/15 (86.7%; 68.8–100.0) | 8/15 (53.3%; 31.2–75.0) |
| 4.5 | Assessment likely influenced by awareness | 11/15 (73.3%; 50.0–92.9) | 7/15 (46.7%; 25.0–69.2) |
| 5.1 | Analysis plan finalised before unblinding | 33/45 (73.3%; 61.0–85.1) | 16/45 (35.6%; 23.1–48.7) |
| 5.2 | Result selected from eligible outcome measurements | 40/45 (88.9%; 80.0–96.1) | 31/44 (70.5%; 57.1–82.9) |
| 5.3 | Result selected from eligible analyses | 34/45 (75.6%; 64.1–86.0) | 32/45 (71.1%; 59.1–82.6) |

SQ, signalling question. Correct collapsed answers / eligible questions (95% CI) on disputed domains. Independent answers define reference; Y/PY, N/PN, NI/NA collapse. Missing predictions are wrong. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 19. Post hoc: supplied documents and answers in question 5.1 losses**

| Trial | Registry, protocol or SAP supplied | Unassisted mentions these documents | Unassisted answer | Adjudicators' answer |
|---|---|---|---|---|
| Allen et al., 2014 | No | No | PY | NI |
| Amini et al., 2022 | No | No | PY | NI |
| Apalla et al., 2013 | No | No | PY | NI |
| Arezzo et al., 2008 | Yes | No | PY | NI |
| ASCET | Yes | No | PY | NI |
| Attal et al., 2016 | Yes | No | PY | NI |
| AWARD-1, 2014 | Yes | No | PY | NI |
| Banovic 2024 (AVATAR) | Yes | No | PY | NI |
| Bente 2014 | No | No | Y | NI |
| De Waal 2021 | Yes | No | PY | NI |
| DURATION-8, 2016 | Yes | No | PY | NI |
| Einarsson 2017 | Yes | No | PY | NI |
| El-Khuffash 2021 | No | No | PY | NI |
| GetGoal-Duo1, 2013 | Yes | No | PY | NI |
| GETGOAL-F1, 2013 | Yes | No | PY | NI |
| Levin 2011 | Yes | No | PY | NI |
| Potsiurko 2022 | Yes | No | PY | NI |
| TESTING | Yes | No | Y | N |

Post hoc description of 18 primary Unassisted losses where question 5.1 alone changed the rating. Supplied means a bundle-inventoried registry, protocol or SAP; relevance/timing are not implied. De Waal's supplied registry is for another trial. Mentions include negated mentions in the Unassisted 5.1 justification. Both adjudicators agreed on each answer. Full justifications and inventories: question51-mechanism.csv. Y, yes; PY, probably yes; NI, no information. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 20. Post hoc: rules-only changes make three arm disagreements ties**

**Changed ratings (post hoc)**

| Trial | Domain | Original | Recomputed |
|---|---|---|---|
| An 2021 | D1 | Some concerns | High |
| Chiba 2021 | D1 | Some concerns | Low |
| Dian 2018 | D1 | Low | Some concerns |
| Gupta 2024 | D1 | Low | High |
| Sung 2020 | D1 | Low | Some concerns |
| Castoldi 2020 | D2 | Some concerns | High |
| Ibrahim 2017 | D2 | Some concerns | Low |
| Stopdapt | D2 | Some concerns | High |
| Albert 2013 | D3 | Some concerns | High |
| Ashar 2022 | D3 | Some concerns | High |
| Levin 2016 | D3 | Some concerns | High |
| Bendix 1996 | D4 | Some concerns | Low |
| Boedt 2023 | D4 | Low | High |
| Fulop 2021 | D4 | Some concerns | Low |
| Abbasi 2012 | D5 | Some concerns | High |
| Aboelfadl 2022 | D5 | Some concerns | High |
| Attal 2021 | D5 | Some concerns | High |
| AWARD-CHN3 2023 | D5 | Low | Some concerns |
| Baillard 2018 | D5 | Low | Some concerns |
| Chiba 2021 | D5 | Some concerns | High |
| Dian 2018 | D5 | Some concerns | High |
| Envision | D5 | Some concerns | High |
| Grant 2017 | D5 | Some concerns | High |
| Jindani 2016 | D5 | Low | Some concerns |
| Stopdapt | D5 | Some concerns | High |

| Trial | Domain | Original | Recomputed |
|---|---|---|---|
| Sung 2020 | D5 | Some concerns | High |

Post hoc. Three of 62 adjudicated arm disagreements changed to ARBITER's preferred rating; 59 were unchanged. Only violations are recomputed from Unassisted answers; invalid paths stay as reported. The original denominator is retained. Three new ties credit both arms, so preferred shares overlap. Weighted κ; exact agreement is n/N (%). CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 20 (continued). Post hoc: rules-only changes make three arm disagreements ties**

**Preferred shares on original adjudicated arm disagreements (post hoc)**

| Version | Measure | Estimate (95% CI) |
|---|---|---|
| Reported | ARBITER | 54/62 (87.1%; 79.0–94.3) |
| Reported | Unassisted | 7/62 (11.3%; 4.5–19.2) |
| Reported | Neither | 1/62 (1.6%; 0.0–4.8) |
| Recomputed | ARBITER | 54/62 (87.1%; 79.0–94.3) |
| Recomputed | Unassisted | 10/62 (16.1%; 8.5–24.6) |
| Recomputed | Neither | 1/62 (1.6%; 0.0–4.8) |

Post hoc. Three of 62 adjudicated arm disagreements changed to ARBITER's preferred rating; 59 were unchanged. Only violations are recomputed from Unassisted answers; invalid paths stay as reported. The original denominator is retained. Three new ties credit both arms, so preferred shares overlap. Weighted κ; exact agreement is n/N (%). CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 20 (continued). Post hoc: rules-only changes make three arm disagreements ties**

**Agreement with published labels (post hoc)**

| Version | Domain | Measure | Estimate (95% CI) |
|---|---|---|---|
| Reported | D1–D5 | Weighted κ | 0.28 (0.21–0.35) |
| Reported | D1–D5 | Exact agreement | 395/510 (77.5%; 74.1–80.8) |
| Reported | D1 | Weighted κ | 0.39 (0.21–0.55) |
| Reported | D1 | Exact agreement | 82/102 (80.4%; 74.5–86.3) |
| Reported | D2 | Weighted κ | 0.06 (−0.10 to 0.21) |
| Reported | D2 | Exact agreement | 73/102 (71.6%; 64.7–78.4) |
| Reported | D3 | Weighted κ | 0.29 (0.13–0.44) |
| Reported | D3 | Exact agreement | 77/102 (75.5%; 68.6–82.4) |
| Reported | D4 | Weighted κ | 0.43 (0.22–0.61) |
| Reported | D4 | Exact agreement | 91/102 (89.2%; 83.3–94.1) |
| Reported | D5 | Weighted κ | 0.25 (0.07–0.43) |
| Reported | D5 | Exact agreement | 72/102 (70.6%; 63.7–78.4) |
| Recomputed | D1–D5 | Weighted κ | 0.24 (0.17–0.31) |
| Recomputed | D1–D5 | Exact agreement | 385/510 (75.5%; 72.0–78.8) |
| Recomputed | D1 | Weighted κ | 0.34 (0.17–0.50) |
| Recomputed | D1 | Exact agreement | 80/102 (78.4%; 71.6–85.3) |
| Recomputed | D2 | Weighted κ | 0.06 (−0.09 to 0.21) |
| Recomputed | D2 | Exact agreement | 73/102 (71.6%; 64.7–78.4) |
| Recomputed | D3 | Weighted κ | 0.26 (0.12–0.41) |
| Recomputed | D3 | Exact agreement | 75/102 (73.5%; 66.7–79.4) |
| Recomputed | D4 | Weighted κ | 0.34 (0.09–0.53) |
| Recomputed | D4 | Exact agreement | 90/102 (88.2%; 83.3–93.1) |
| Recomputed | D5 | Weighted κ | 0.17 (0.02–0.33) |
| Recomputed | D5 | Exact agreement | 67/102 (65.7%; 57.8–73.5) |

Post hoc. Three of 62 adjudicated arm disagreements changed to ARBITER's preferred rating; 59 were unchanged. Only violations are recomputed from Unassisted answers; invalid paths stay as reported. The original denominator is retained. Three new ties credit both arms, so preferred shares overlap. Weighted κ; exact agreement is n/N (%). CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 20 (continued). Post hoc: rules-only changes make three arm disagreements ties**

**Agreement with the adjudicators' agreed choice (post hoc)**

| Version | Estimate (95% CI) |
|---|---|
| Reported | 69/129 (53.5%; 46.0–60.7) |
| Recomputed | 64/129 (49.6%; 41.7–57.4) |

Post hoc. Three of 62 adjudicated arm disagreements changed to ARBITER's preferred rating; 59 were unchanged. Only violations are recomputed from Unassisted answers; invalid paths stay as reported. The original denominator is retained. Three new ties credit both arms, so preferred shares overlap. Weighted κ; exact agreement is n/N (%). CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 21. RoB 2 consistency and structural audit, all sources and domains**

| Domain | Measure | ARBITER (Sonnet 5.5) | Unassisted (Sonnet 5.5) | Published |
|---|---|---|---|---|
| Overall | Consistent with RoB 2 | 101/102 (99.0%) | 102/102 (100.0%) | 74/81 (91.4%) |
| Overall | Permitted escalation | 1/102 (1.0%) | 0/102 (0.0%) | 6/81 (7.4%) |
| Overall | Rating contradicts its own answers | 0/102 (0.0%) | 0/102 (0.0%) | 1/81 (1.2%) |
| D1–D5 | Consistent with RoB 2 | 510/510 (100.0%) | 444/509 (87.2%) | — |
| D1–D5 | Proposed judgement matches algorithm | 510/510 (100.0%) | — | — |
| D1–D5 | Recorded override | 0/510 (0.0%) | — | — |
| D1–D5 | Rating contradicts its own answers | 0/510 (0.0%) | 26/509 (5.1%) | — |
| D1–D5 | Required question skipped | 0/510 (0.0%) | 39/509 (7.7%) | — |
| D1–D5 | Any rule break | 0/510 (0.0%) | 65/509 (12.8%) | — |
| D1 | Consistent with RoB 2 | 102/102 (100.0%) | 96/101 (95.0%) | — |
| D1 | Proposed judgement matches algorithm | 102/102 (100.0%) | — | — |
| D1 | Recorded override | 0/102 (0.0%) | — | — |
| D1 | Rating contradicts its own answers | 0/102 (0.0%) | 5/101 (5.0%) | — |
| D1 | Required question skipped | 0/102 (0.0%) | 0/101 (0.0%) | — |
| D1 | Any rule break | 0/102 (0.0%) | 5/101 (5.0%) | — |
| D2 | Consistent with RoB 2 | 102/102 (100.0%) | 95/102 (93.1%) | — |
| D2 | Proposed judgement matches algorithm | 102/102 (100.0%) | — | — |
| D2 | Recorded override | 0/102 (0.0%) | — | — |
| D2 | Rating contradicts its own answers | 0/102 (0.0%) | 3/102 (2.9%) | — |
| D2 | Required question skipped | 0/102 (0.0%) | 4/102 (3.9%) | — |
| D2 | Any rule break | 0/102 (0.0%) | 7/102 (6.9%) | — |
| D3 | Consistent with RoB 2 | 102/102 (100.0%) | 97/102 (95.1%) | — |
| D3 | Proposed judgement matches algorithm | 102/102 (100.0%) | — | — |
| D3 | Recorded override | 0/102 (0.0%) | — | — |
| D3 | Rating contradicts its own answers | 0/102 (0.0%) | 3/102 (2.9%) | — |
| D3 | Required question skipped | 0/102 (0.0%) | 2/102 (2.0%) | — |
| D3 | Any rule break | 0/102 (0.0%) | 5/102 (4.9%) | — |

| Domain | Measure | ARBITER (Sonnet 5.5) | Unassisted (Sonnet 5.5) | Published |
|---|---|---|---|---|
| D4 | Consistent with RoB 2 | 102/102 (100.0%) | 67/102 (65.7%) | — |
| D4 | Proposed judgement matches algorithm | 102/102 (100.0%) | — | — |
| D4 | Recorded override | 0/102 (0.0%) | — | — |
| D4 | Rating contradicts its own answers | 0/102 (0.0%) | 3/102 (2.9%) | — |
| D4 | Required question skipped | 0/102 (0.0%) | 32/102 (31.4%) | — |
| D4 | Any rule break | 0/102 (0.0%) | 35/102 (34.3%) | — |
| D5 | Consistent with RoB 2 | 102/102 (100.0%) | 89/102 (87.3%) | — |
| D5 | Proposed judgement matches algorithm | 102/102 (100.0%) | — | — |
| D5 | Recorded override | 0/102 (0.0%) | — | — |
| D5 | Rating contradicts its own answers | 0/102 (0.0%) | 12/102 (11.8%) | — |
| D5 | Required question skipped | 0/102 (0.0%) | 1/102 (1.0%) | — |
| D5 | Any rule break | 0/102 (0.0%) | 13/102 (12.7%) | — |
| Audit | Answers citing ≥1 supporting passage or image | 1538/1538 (100.0%) | — | — |
| Audit | Audit file passes the independent verifier | 102/102 (100.0%) | — | — |
| Audit | All evidence bound | 1421/1421 (100.0%) | — | — |
| Audit | Quoted text linked to its source | 1407/1407 (100.0%) | — | — |
| Audit | Image regions linked to their page | 14/14 (100.0%) | — | — |
| Audit | Other evidence records | 0/1421 (0.0%) | — | — |
| Audit | Trials using visual evidence | 8/102 (7.8%) | — | — |

Values are n/N (%). Proposed judgements determine ARBITER domain consistency; adopted overrides are separate. Overall exact worst-domain agreement and permitted escalation are separate. Evidence coverage uses active signalling answers as the denominator and answers citing at least one evidence record as the numerator. Unassisted and Published evidence traces are unavailable. Binding and independent verification are structural checks, not clinical correctness. Visual transcriptions are model-written and not checked against images. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 22. Required question skipped sensitivity**

| Domain | Evaluable (n) | Original violations (n) | Invalid paths (n) | Additional violations (n) | Total violations (n) |
|---|---|---|---|---|---|
| **Unassisted (Sonnet 5.5) — replacement: N/PN** | | | | | |
| D1–D5 | 509 | 26 | 39 | 3 | 29 |
| D1 | 101 | 5 | 0 | 0 | 5 |
| D2 | 102 | 3 | 4 | 1 | 4 |
| D3 | 102 | 3 | 2 | 2 | 5 |
| D4 | 102 | 3 | 32 | 0 | 3 |
| D5 | 102 | 12 | 1 | 0 | 12 |
| **Unassisted (Sonnet 5.5) — replacement: NI** | | | | | |
| D1–D5 | 509 | 26 | 39 | 36 | 62 |
| D1 | 101 | 5 | 0 | 0 | 5 |
| D2 | 102 | 3 | 4 | 2 | 5 |
| D3 | 102 | 3 | 2 | 2 | 5 |
| D4 | 102 | 3 | 32 | 32 | 35 |
| D5 | 102 | 12 | 1 | 0 | 12 |

Required invalid answers recursively resolve as N/PN then NI, including newly required questions. No off-path imputation; these are counts, not new assessments. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 23. Quotation syntax in active justifications**

| Source | Estimate (95% CI) |
|---|---|
| ARBITER (Sonnet 5.5) | 34/1538 (2.2%; 1.1–3.5) |
| Unassisted (Sonnet 5.5) | 0/1459 (0.0%) |

Quoted active answers / all active answers. Paired double quotes enclose at least five words. Syntax is not verification and understates ARBITER bundle-based grounding. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 24. Unassisted justification content**

| Source | Answer-code-only / exported fields (%) |
|---|---|
| Unassisted (Sonnet 5.5) | 1344/1459 (92.1%) |

Codes are Y, PY, PN, N, NI or NA. This describes exported fields without inferring absent reasoning. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 25. Time, cost and token use per trial by model**

| Model | Metric | ARBITER | Unassisted | Paired mean difference (95% CI) |
|---|---|---|---|---|
| Sonnet 5.5 | System runtime (min/trial) | 7.6 [6.1–10.5] | 0.5 [0.4–0.5] | 11.0 (8.4–14.1) |
| Sonnet 5.5 | API list-price equivalent (US$/trial) | 3.78 [3.46–4.10] | 0.09 [0.07–0.13] | 3.68 (3.61–3.75) |
| Sonnet 5.5 | Input tokens (count/trial) | 96 [86–104] | 6 [6–6] | 90 (88–92) |
| Sonnet 5.5 | Cache creation tokens (millions/trial) | 0.49 [0.46–0.54] | 0.02 [0.01–0.03] | 0.48 (0.47–0.49) |
| Sonnet 5.5 | Cache read tokens (millions/trial) | 4.96 [4.48–5.59] | 0.02 [0.02–0.02] | 4.99 (4.87–5.12) |
| Sonnet 5.5 | Output tokens (thousands/trial) | 21.4 [18.9–24.3] | 2.4 [1.9–2.9] | 19.5 (18.8–20.1) |
| Sonnet 5.5 | Thinking tokens (thousands/trial) | 3.5 [2.6–4.2] | 1.0 [0.7–1.3] | 2.5 (2.3–2.7) |
| Sonnet 5.5 | Agent steps/trial | 60.0 [54.0–71.8] | 4.0 [4.0–5.0] | 58.6 (56.9–60.3) |
| Sonnet 5.5 | Continue prompts/trial | 0.0 [0.0–0.0] | 0.0 [0.0–0.0] | 0.0 (0.0–0.0) |
| Sonnet 5.5 | Turns/trial | 2.0 [2.0–2.0] | 1.0 [1.0–1.0] | 1.0 (1.0–1.0) |

102 trials per arm; per-trial median [IQR] and paired mean ARBITER minus Unassisted difference (95% CI). US$ denotes API list-price equivalent; runs used a subscription. Thinking is included in output; eMethods give the reconstruction formula and rates. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 26. API cost reconstruction: estimates and bounds**

| Measure | ARBITER (Sonnet 5.5) | Unassisted (Sonnet 5.5) |
|---|---|---|
| Primary estimate: total (USD) | 386.07 | 10.49 |
| Primary estimate: median per trial (USD) | 3.78 | 0.09 |
| Lower reconstruction bound: total (USD) | 386.07 | 10.49 |
| Lower bound: median per trial (USD) | 3.78 | 0.09 |
| Upper reconstruction bound: total (USD) | 386.07 | 10.49 |
| Upper bound: median per trial (USD) | 3.78 | 0.09 |
| Harness diagnostic: total (USD) | 1140.34 | 27.12 |
| Harness diagnostic: median per trial (USD) | 11.11 | 0.24 |
| 5-minute write sensitivity: total (USD) | 365.88 | 10.49 |
| 5-minute write sensitivity: median per trial (USD) | 3.58 | 0.09 |
| 5-minute sensitivity lower bound: total (USD) | 365.88 | 10.49 |
| 5-minute sensitivity lower bound: median per trial (USD) | 3.58 | 0.09 |
| 5-minute sensitivity upper bound: total (USD) | 365.88 | 10.49 |
| 5-minute sensitivity upper bound: median per trial (USD) | 3.58 | 0.09 |

USD. Reconstruction bounds are not sampling CIs. Five-minute side-call cache writes are a sensitivity. Harness figures are unusable: unrecognised model and unknown cost basis. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 27. Token reconciliation and compaction diagnostics**

| Measure | ARBITER (Sonnet 5.5) | Unassisted (Sonnet 5.5) |
|---|---|---|
| Main-thread request input tokens (n) | 9,794 | 644 |
| Main-thread request 1-hour cache write tokens (n) | 50,698,919 | 1,910,531 |
| Main-thread request 5-minute cache write tokens (n) | 0 | 0 |
| Main-thread request cache read tokens (n) | 512,761,030 | 3,585,005 |
| Terminal input tokens (n) | 9,794 | 644 |
| Terminal 1-hour cache write tokens (n) | 50,695,878 | 1,910,531 |
| Terminal 5-minute cache write tokens (n) | 0 | 0 |
| Terminal cache read tokens (n) | 512,867,801 | 3,585,005 |
| Terminal output tokens (n) | 2,237,043 | 248,579 |
| Side-call input tokens (n) | 5,777,447 | 0 |
| Side-call 1-hour cache write tokens (n) | 13,464,429 | 0 |
| Side-call 5-minute cache write tokens (n) | 0 | 0 |
| Side-call cache read tokens (n) | 203,137,779 | 0 |
| Side-call output tokens (n) | 2,388,538 | 0 |
| Terminal minus request input tokens (n) | 0 | 0 |
| Terminal minus request 1-hour cache write tokens (n) | −3,041 | 0 |
| Terminal minus request 5-minute cache write tokens (n) | 0 | 0 |
| Terminal minus request cache read tokens (n) | 106,771 | 0 |
| Main-thread requests (n) | 4,897 | 322 |
| Requests with prompts >100,000 tokens (n) | 3,272 | 0 |
| Compaction events (n) | 322 | 0 |
| Turns with side calls (n) | 122 | 0 |
| Turns with request counts above terminal counts (n) | 3 | 0 |
| Side-call turns without compaction (n) | 0 | 0 |

Token totals across trials. Terminal minus request is signed, including retry residuals. Side calls are total model usage minus terminal usage; output already includes thinking. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 28. Evidence and quotation locations for the prior-work comparison**

| Study | Quoted cell evidence and locations | Source |
|---|---|---|
| [ref: Marshall 2016] | Code rules — No; Enforcement — No; Agent — No: “Support Vector Machine” — Methods, Machine-Learning Approach: learned document/sentence classifiers, not official decision-rule execution or a tool-calling agent. Three classifiers use held-out trial reports; model internals differ.<br>Registry/protocol — No: “Linking with this dataset may yield better performance.” — Discussion, selective reporting limitation: protocol/registry linkage is future work; current models use trial-publication text.<br>Identical inputs — Yes; Fixed-model comparison — No: “test data” — Evaluation, Task 1: the held-out duplicate-assessment trial set is used across three classifier variants. This compares different classifiers, not the same model with/without an assessment workflow.<br>Reference; Independent reference — Partly: “blinded expert panel” — Introduction and Evaluation: historical independent Cochrane assessments supply RoB labels; newly recruited blinded experts assess supporting-text relevance only. No newly adjudicated RoB reference is implied. | PMC4713900, Methods and Evaluation. |
| [ref: Eisele-Metzger 2025] | Rule checks — Partly: “checked the 100 overall judgements” — Methods, Data analysis: overall-rule check only; 2/100 violations; not signalling-answer/domain conformance.<br>Independent reference — Partly: “We reviewed all ‘two-level discrepancies’” — Methods, Data analysis: unblinded author review against sources, not an independent panel or replacement reference. | PMC12527486, Methods. |
| [ref: Huang 2025] | Code rules — Partly: “using Excel based on the RoB2 decision algorithm” — Discussion, Principal Findings: secondary spreadsheet derivation; executable formulas/automation are not established.<br>Rule checks — Partly: “investigated the discrepancies” — Methods, Statistical Analysis: direct versus algorithm-derived judgments compared; no violation count/rate.<br>Independent reference — Yes: “initially conducted independent judgments” — Methods, Establishment of the Criterion Standard: three trained reviewers, then consensus.<br>Registry/protocol — No: “excluding Multimedia Appendices and protocols” — Discussion, Principal Findings; Limitations additionally excludes registration documents. Protocol availability is metadata, not delivery of a protocol. | PMC12238788, Methods and Discussion. |
| [ref: Lai 2026] | Independent reference — Partly: “manually corrected based on research team consensus” — Methods, Establishment of the Reference Standard; Results sensitivity analysis: corrections to Toouli assessments, not a new blinded reference. | PMC13614218, Methods and Results. |
| [ref: Nyrhi 2026] | Registry/protocol — Yes: “PDF-versions of their protocols when available” — Methods, Prompt development and standardisation: available protocols supplied; DeepSeek received copied protocol text (eAppendix 1). Missing protocols for some trials do not turn documented delivery into Partly; the column does not require a protocol for every trial. | PMC13054287, Methods; eAppendix 1. |
| [ref: Gandhi 2026] | Agent — Partly: “web-browsing capabilities enabled” — Methods, Data collection and RoB assessment: capability enabled; actual external searches not logged/quantified.<br>Independent reference — Yes: “masked to the OSRA’s assessments” — Same section: three reviewers independently re-rated all 50 trials; consensus/fourth reviewer resolved their differences. | PMC13334998, Methods. |

| Study | Quoted cell evidence and locations | Source |
|---|---|---|
| [ref: Choi 2026] | Code rules — Yes: “implemented in code” — Methods, RoB 2.0 Algorithm Implementation: authors recomputed domain judgments.<br>Enforcement — Partly: “official flowchart aggregation logic” — Methods, Study Design and Data: ROBoto2 component only; general Gemini/Gems emit ratings.<br>Rule checks — Yes: “conformance was 43%” — Results, Algorithm Conformance and Internal Consistency: 43%, 51%, 60% across configurations; skip logic also checked.<br>Agent — Partly: “reference knowledge” — Methods, Study Design and Data: Gems retrieval and ROBoto2 pipeline, not demonstrated autonomous tool calls.<br>Independent reference — Yes: “One experienced reviewer’s assessment served as the reference.” — Methods, Study Design and Data: a single expert, not a blinded panel. | Publisher full article, Methods and Results. |
| [ref: Hevia 2025] | Code rules — Yes; Enforcement — Partly: “implement the logic provided by the ROB2 flowcharts” — Section 4, Domain-level and overall judgments: integrated software aggregation; not proven rejection of invalid/incomplete answers, and Choi found displayed inconsistencies.<br>Audit file — Partly: “retains the original LLM responses and rationales” — Section 4, Collecting user feedback: original and human-edited records retained; no independently verified source-integrity bundle.<br>Agent — Partly: “retrieved evidence paragraphs as context” — Section 4, Answering signaling questions: retrieval pipeline, not autonomous agent tool selection.<br>Independent reference — Yes: “245 reports” — Section 5, Annotation procedure: manually assessed expert subset used for every LLM evaluation. | arXiv:2511.03048 v1, sections 4–5. |
| [ref: Liang 2026] | Quotation checks — Partly: “a verbatim quotation drawn from the retrieved trial text” — Methods, Retrieval-Based Evidence Support and Hallucination Assessment: GPT-5 judge supplies quoted support verdicts; no deterministic occurrence/hash check established.<br>Audit file — Partly: “returned structured JSON” — Same section: verdict, quote and rationale are reviewable; no complete independently verifiable source bundle established.<br>Agent — Partly: “post hoc claim verification” — Discussion, Limitations: BM25 plus judge pipeline evaluates generated claims; no autonomous assessment agent established. | PMC13628037, Methods and Discussion. |
| [ref: Saadi 2026] | Agent — Not reported: “customised agent-based implementations” — Methods, LLM assessment workflow: structured prompting is established, but neither retrieval nor external tool calling is demonstrated.<br>Independent reference — Partly: “two separate teams of human reviewers” — Methods, Study design and data source: concordant historical review ratings selected, not new adjudication. | PMC13528376, Methods. |

| Study | Quoted cell evidence and locations | Source |
|---|---|---|
| This study (ARBITER) | Code rules — Yes: “ARBITER's server proposes the algorithm judgement.” — ANALYSIS-PLAN, section 5.4, resolution dated 18 October 2026.<br>Enforcement — Yes: “The assessing model may adopt a different judgement only through a reasoned override” — Same resolution: enforced proposal/recording path, not a claim that overrides are impossible.<br>Rule checks — Yes: “zero domain rule breaks by design” — Results, rule consistency; Table 3, ARBITER D1–D5 rule-break row (0/510).<br>Quotation checks — Yes: “every quotation is found in a stored source” — ANALYSIS-PLAN, section 7, Resolution; audit-summary.csv reports 1407/1407 text-span and 14/14 visual evidence bindings.<br>Audit file — Yes: “containing `canonical.json` and `verification.json`” — ANALYSIS-PLAN, section 5.8, bundle definition; audit-summary.csv reports 102/102 bundle verification.<br>Agent — Yes: “its MCP server.” — ARBITER run input inventory, Setup.<br>Independent reference — Yes: “Blinded dual adjudication of the disagreement cells” — ANALYSIS-PLAN, section 7, Action: adjudicators first rated independently, then chose among blinded labels.<br>Registry/protocol — Yes: “the 60 protocol and SAP PDFs” — Unassisted run input inventory, Inputs; ARBITER run input inventory, opening paragraph: registry document acquisition enabled. Availability varies by trial; Yes records delivery of the complete acquired ancillary set to both arms, not that every trial had a protocol. | This study, Methods, Results and Table 3. |

One row per Table 4 study; reference callouts resolve in the bibliography. Quotations condense documented feature decisions and their locations. NR means not reported, not absent; definitions are in Table 4. Text-relevance review is not risk-of-bias adjudication. Quotation occurrence does not establish support. Recorded overrides were permitted; none occurred in 510 domains. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet’s linearly weighted agreement coefficient; κ, Cohen’s kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 29. Agreement against the composite reference, Sonnet only**

| Domain | Weighted κ | AC2 | Exact agreement |
|---|---|---|---|
| **ARBITER (Sonnet 5.5)** | | | |
| D1–D5 | 0.94 (0.91–0.97) | 0.98 (0.97–0.99) | 472/485 (97.3%; 95.9–98.6) |
| D1 | 0.97 (0.92–1.00) | 0.99 (0.98–1.00) | 97/98 (99.0%; 96.9–100.0) |
| D2 | 0.80 (0.61–0.93) | 0.96 (0.93–0.99) | 88/94 (93.6%; 88.7–97.9) |
| D3 | 0.97 (0.92–1.00) | 0.98 (0.96–1.00) | 92/94 (97.9%; 94.7–100.0) |
| D4 | 0.92 (0.80–1.00) | 0.99 (0.97–1.00) | 99/101 (98.0%; 95.1–100.0) |
| D5 | 0.96 (0.91–1.00) | 0.98 (0.96–1.00) | 96/98 (98.0%; 95.8–100.0) |
| Post hoc split: Adjudicators' agreed choice | — | — | 116/129 (89.9%) |
| Post hoc split: Kept published label | — | — | 356/356 (100.0%) |
| **Unassisted (Sonnet 5.5)** | | | |
| D1–D5 | 0.69 (0.62–0.76) | 0.91 (0.88–0.93) | 425/485 (87.6%; 85.2–90.0) |
| D1 | 0.89 (0.77–1.00) | 0.97 (0.94–1.00) | 95/98 (96.9%; 93.8–100.0) |
| D2 | 0.42 (0.14–0.65) | 0.88 (0.82–0.92) | 78/94 (83.0%; 76.7–89.1) |
| D3 | 0.78 (0.65–0.89) | 0.90 (0.85–0.95) | 83/94 (88.3%; 82.3–93.8) |
| D4 | 0.69 (0.48–0.86) | 0.96 (0.93–0.98) | 94/101 (93.1%; 88.2–97.0) |
| D5 | 0.55 (0.44–0.67) | 0.80 (0.74–0.86) | 75/98 (76.5%; 69.7–83.2) |
| Post hoc split: Adjudicators' agreed choice | — | — | 69/129 (53.5%) |
| Post hoc split: Kept published label | — | — | 356/356 (100.0%) |
| **All-Low rule** | | | |
| D1–D5 | — | — | 358/485 (73.8%; 70.6–77.0) |
| D1 | — | — | 78/98 (79.6%; 73.5–85.7) |
| D2 | — | — | 79/94 (84.0%; 77.7–90.4) |
| D3 | — | — | 66/94 (70.2%; 63.5–76.8) |
| D4 | — | — | 87/101 (86.1%; 80.4–91.2) |
| D5 | — | — | 48/98 (49.0%; 41.7–56.6) |

Weighted κ, AC2 and exact agreement. The composite uses the agreed choice on 129 cells and Published on 356; 25 cells where adjudicators split are excluded. Adjudicators chose among three labels; unreviewed cells matching Published are assumed correct. Both inflate absolute values. Post hoc component rows give n/N (%) without new intervals. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 30. Paired differences against the composite reference**

| Domain | Weighted κ | AC2 | Exact agreement |
|---|---|---|---|
| **ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5)** | | | |
| D1–D5 | 0.25 (0.18–0.33) | 0.07 (0.05–0.10) | 9.7 (7.0–12.4) |
| D1 | 0.08 (0.00–0.20) | 0.02 (0.00–0.05) | 2.0 (0.0–5.1) |
| D2 | 0.38 (0.12–0.68) | 0.08 (0.03–0.14) | 10.6 (3.2–18.1) |
| D3 | 0.18 (0.07–0.32) | 0.08 (0.03–0.14) | 9.6 (3.4–15.8) |
| D4 | 0.23 (0.00–0.48) | 0.03 (0.00–0.06) | 5.0 (0.0–10.0) |
| D5 | 0.41 (0.28–0.52) | 0.18 (0.12–0.24) | 21.4 (14.3–28.3) |

ARBITER minus Unassisted; coefficients in original units, proportions in percentage points. Reference: the adjudicators' agreed choice where both agreed; published labels elsewhere. Excludes 25 divided decisions (485 cells per arm). Only three labels were offered. Unreviewed cells where both arms matched published labels are assumed correct. Both features inflate absolute values. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 31. Composite-reference sensitivity estimates by adjudicator**

| Adjudicator | Source | Domain | Weighted κ | AC2 | Exact agreement |
|---|---|---|---|---|---|
| Adjudicator 1 | ARBITER (Sonnet 5.5) | D1–D5 | 0.86 (0.81–0.90) | 0.95 (0.93–0.97) | 479/510 (93.9%; 92.0–95.9) |
| Adjudicator 1 | ARBITER (Sonnet 5.5) | D1 | 0.90 (0.81–0.98) | 0.97 (0.95–0.99) | 98/102 (96.1%; 93.1–99.0) |
| Adjudicator 1 | ARBITER (Sonnet 5.5) | D2 | 0.70 (0.50–0.85) | 0.92 (0.87–0.96) | 92/102 (90.2%; 85.3–95.1) |
| Adjudicator 1 | ARBITER (Sonnet 5.5) | D3 | 0.83 (0.72–0.92) | 0.90 (0.84–0.95) | 92/102 (90.2%; 85.3–95.1) |
| Adjudicator 1 | ARBITER (Sonnet 5.5) | D4 | 0.92 (0.80–1.00) | 0.99 (0.97–1.00) | 100/102 (98.0%; 95.1–100.0) |
| Adjudicator 1 | ARBITER (Sonnet 5.5) | D5 | 0.90 (0.82–0.96) | 0.96 (0.92–0.98) | 97/102 (95.1%; 91.2–98.0) |
| Adjudicator 1 | Unassisted (Sonnet 5.5) | D1–D5 | 0.65 (0.58–0.72) | 0.88 (0.85–0.90) | 435/510 (85.3%; 82.5–87.8) |
| Adjudicator 1 | Unassisted (Sonnet 5.5) | D1 | 0.87 (0.75–0.98) | 0.97 (0.93–0.99) | 98/102 (96.1%; 92.2–99.0) |
| Adjudicator 1 | Unassisted (Sonnet 5.5) | D2 | 0.35 (0.12–0.55) | 0.82 (0.76–0.88) | 79/102 (77.5%; 70.6–84.3) |
| Adjudicator 1 | Unassisted (Sonnet 5.5) | D3 | 0.71 (0.57–0.83) | 0.85 (0.78–0.92) | 87/102 (85.3%; 79.4–91.2) |
| Adjudicator 1 | Unassisted (Sonnet 5.5) | D4 | 0.68 (0.50–0.84) | 0.95 (0.92–0.98) | 94/102 (92.2%; 87.3–96.1) |
| Adjudicator 1 | Unassisted (Sonnet 5.5) | D5 | 0.54 (0.42–0.66) | 0.79 (0.73–0.85) | 77/102 (75.5%; 68.6–82.4) |
| Adjudicator 1 | All-Low rule | D1–D5 | — | — | 364/510 (71.4%; 67.8–74.9) |
| Adjudicator 1 | All-Low rule | D1 | — | — | 79/102 (77.5%; 70.6–84.3) |
| Adjudicator 1 | All-Low rule | D2 | — | — | 81/102 (79.4%; 72.5–86.3) |
| Adjudicator 1 | All-Low rule | D3 | — | — | 69/102 (67.6%; 60.8–74.5) |
| Adjudicator 1 | All-Low rule | D4 | — | — | 87/102 (85.3%; 79.4–91.2) |
| Adjudicator 1 | All-Low rule | D5 | — | — | 48/102 (47.1%; 40.2–54.9) |
| Adjudicator 2 | ARBITER (Sonnet 5.5) | D1–D5 | 0.91 (0.88–0.94) | 0.97 (0.96–0.98) | 488/510 (95.7%; 94.1–97.1) |
| Adjudicator 2 | ARBITER (Sonnet 5.5) | D1 | 0.95 (0.87–1.00) | 0.99 (0.97–1.00) | 100/102 (98.0%; 95.1–100.0) |
| Adjudicator 2 | ARBITER (Sonnet 5.5) | D2 | 0.71 (0.52–0.85) | 0.93 (0.88–0.96) | 91/102 (89.2%; 83.3–94.1) |
| Adjudicator 2 | ARBITER (Sonnet 5.5) | D3 | 0.96 (0.90–1.00) | 0.97 (0.95–1.00) | 99/102 (97.1%; 94.1–100.0) |
| Adjudicator 2 | ARBITER (Sonnet 5.5) | D4 | 0.89 (0.78–1.00) | 0.98 (0.96–1.00) | 99/102 (97.1%; 94.1–100.0) |

| Adjudicator | Source | Domain | Weighted κ | AC2 | Exact agreement |
|---|---|---|---|---|---|
| Adjudicator 2 | ARBITER (Sonnet 5.5) | D5 | 0.94 (0.88–1.00) | 0.97 (0.95–1.00) | 99/102 (97.1%; 94.1–100.0) |
| Adjudicator 2 | Unassisted (Sonnet 5.5) | D1–D5 | 0.67 (0.59–0.73) | 0.89 (0.87–0.91) | 439/510 (86.1%; 83.3–88.6) |
| Adjudicator 2 | Unassisted (Sonnet 5.5) | D1 | 0.82 (0.68–0.94) | 0.95 (0.91–0.99) | 96/102 (94.1%; 89.2–98.0) |
| Adjudicator 2 | Unassisted (Sonnet 5.5) | D2 | 0.50 (0.25–0.70) | 0.87 (0.82–0.92) | 85/102 (83.3%; 77.5–89.2) |
| Adjudicator 2 | Unassisted (Sonnet 5.5) | D3 | 0.72 (0.58–0.83) | 0.86 (0.80–0.92) | 86/102 (84.3%; 78.4–90.2) |
| Adjudicator 2 | Unassisted (Sonnet 5.5) | D4 | 0.73 (0.55–0.88) | 0.96 (0.93–0.98) | 95/102 (93.1%; 88.2–97.1) |
| Adjudicator 2 | Unassisted (Sonnet 5.5) | D5 | 0.52 (0.40–0.65) | 0.79 (0.73–0.85) | 77/102 (75.5%; 68.6–82.4) |
| Adjudicator 2 | All-Low rule | D1–D5 | — | — | 372/510 (72.9%; 69.6–76.1) |
| Adjudicator 2 | All-Low rule | D1 | — | — | 81/102 (79.4%; 73.5–85.3) |
| Adjudicator 2 | All-Low rule | D2 | — | — | 84/102 (82.4%; 76.5–88.2) |
| Adjudicator 2 | All-Low rule | D3 | — | — | 68/102 (66.7%; 60.8–72.5) |
| Adjudicator 2 | All-Low rule | D4 | — | — | 87/102 (85.3%; 79.4–91.2) |
| Adjudicator 2 | All-Low rule | D5 | — | — | 52/102 (51.0%; 44.1–58.8) |

510 cells per Sonnet arm. Each adjudicator's selected label replaces Published on disputed cells. Only three labels were offered; unadjudicated cells are assumed correct. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

**eTable 32. Composite-reference paired differences by adjudicator**

| Adjudicator | Comparison | Domain | Weighted κ | AC2 | Exact agreement |
|---|---|---|---|---|---|
| Adjudicator 1 | ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5) | D1–D5 | 0.21 (0.14–0.29) | 0.07 (0.04–0.09) | 8.6 (5.7–11.6) |
| Adjudicator 1 | ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5) | D1 | 0.03 (−0.09 to 0.16) | 0.01 (−0.02 to 0.04) | 0.0 (−3.9 to 3.9) |
| Adjudicator 1 | ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5) | D2 | 0.35 (0.11–0.62) | 0.09 (0.04–0.15) | 12.7 (4.9–20.6) |
| Adjudicator 1 | ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5) | D3 | 0.12 (−0.02 to 0.26) | 0.05 (−0.02 to 0.12) | 4.9 (−2.0 to 11.8) |
| Adjudicator 1 | ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5) | D4 | 0.24 (0.02–0.46) | 0.04 (0.01–0.07) | 5.9 (1.0–11.8) |
| Adjudicator 1 | ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5) | D5 | 0.36 (0.22–0.50) | 0.17 (0.10–0.23) | 19.6 (11.8–27.5) |
| Adjudicator 2 | ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5) | D1–D5 | 0.24 (0.17–0.32) | 0.08 (0.05–0.10) | 9.6 (6.9–12.5) |
| Adjudicator 2 | ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5) | D1 | 0.13 (0.03–0.25) | 0.03 (0.01–0.07) | 3.9 (1.0–7.8) |
| Adjudicator 2 | ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5) | D2 | 0.21 (−0.03 to 0.47) | 0.05 (−0.01 to 0.11) | 5.9 (−2.9 to 13.7) |
| Adjudicator 2 | ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5) | D3 | 0.24 (0.12–0.38) | 0.12 (0.05–0.18) | 12.7 (5.9–19.6) |
| Adjudicator 2 | ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5) | D4 | 0.16 (−0.06 to 0.39) | 0.02 (−0.01 to 0.06) | 3.9 (−1.0 to 8.8) |
| Adjudicator 2 | ARBITER (Sonnet 5.5) minus Unassisted (Sonnet 5.5) | D5 | 0.42 (0.28–0.54) | 0.18 (0.12–0.24) | 21.6 (14.7–28.4) |

ARBITER minus Unassisted, coefficient units or percentage points; 510 paired cells per arm. Only offered labels could be chosen; unadjudicated cells are assumed correct. CI, confidence interval; 10,000 percentile bootstrap replicates, trials resampled within meta-analysis; seed 20261018. CI, confidence interval; IQR, interquartile range; USD, US dollars; AC2, Gwet's linearly weighted agreement coefficient; κ, Cohen's kappa (linear weights unless explicitly unweighted); RoB 2, risk of bias tool for randomised trials; D1–D5, its five domains; Y/PY, yes/probably yes; N/PN, no/probably no; NI, no information; NA, not applicable. All counts are numbers of the units named in the headers; — means undefined or not applicable.

## Prompt templates and a filled example

### ARBITER template

/arbiter-assess Assess risk of bias for {outcome} defined as {definition} in {trial}.

### ARBITER example

/arbiter-assess Assess risk of bias for Adverse events graded by CTCAE v4.0; 'Grade 3 (severe AEs)' row of Table 3; patients with grade 3 events in Table 4 defined as Table 3 Grade 3 (severe AEs) number of AEs: 600 mg 6, 900 mg 5, 1,200 mg 9 (all subjects 20); Grade 4: 0; Grade 5 (death): 1 per arm. Table 4 total no. of patients with grade 3 events: 2, 4, 5. Abstract: 'Grade 3 adverse events occurred in only 2 patients in the 600-mg arm, 4 patients in the 900-mg arm, and 5 patients in the 1,200-mg arm.' in Aarnoutse 2017.

### Unassisted template

Assess risk of bias for {outcome} defined as {definition} in {trial}.

### Unassisted output-format instruction

When you have finished, give your answers in this format: a table with one row per signalling question (columns: domain, question number, answer as Y, PY, PN, N, NI or NA), and then a final line exactly like "FINAL: D1=<judgement>; D2=<judgement>; D3=<judgement>; D4=<judgement>; D5=<judgement>; Overall=<judgement>", where each judgement is Low, Some concerns or High.

## Run provenance

Model claude-sonnet-5-5; effort medium; Claude Code 2.1.278; isolated configuration homes. ARBITER 1.0.0, public commit 2f549ae30f056ff441ea61a04b719968716f2219. ARBITER with Sonnet ran 12 October 2026 (12:12-19:17 EDT); other arms ran 14 October 2026. Harness approved the proposed result unchanged and replied Approved.; uncoached Continue. replies were permitted at most twice per phase. Each trial ran once in each arm; run-to-run variability is not estimated. Unassisted access was limited to Read, Glob and Grep, without an MCP server, skills or web tools. Supplied registry Markdown and acquired protocol/SAP documents preserved input parity. Trial PDFs are not redistributed; source DOIs and FILES.csv hashes identify them. Bootstrap seed 20261018, 10,000 replicates; R/renv/Quarto versions are recorded in the project lockfile and session-info.txt.